

Al-Muneer Online Portal

Event Venue Management Systems - Web Development

AHMED HANI GHALEB

UNIVERSITI TEKNOLOGI MALAYSIA

UNIVERSITI TEKNOLOGI MALAYSIA

**DECLARATION OF THESIS / UNDERGRADUATE PROJECT REPORT AND
COPYRIGHT**

Author's full name Ahmed Hani Ahmed Ghaleb:

Date of Birth : 2001/04/19

Title : Al-Muneer Hall Reservations

Academic Session : 24/25 2

I declare that this thesis is classified as:

I acknowledged that Universiti Teknologi Malaysia reserves the right as follows:

- The thesis is the property of Universiti Teknologi Malaysia
- The Library of Universiti Teknologi Malaysia has the right to make copies for the purpose of research only.
- The Library has the right to make copies of the thesis for academic exchange.

NOTES : If the thesis is CONFIDENTIAL or RESTRICTED, please attach with the letter from the organization with period and reasons for confidentiality or restriction

“I hereby declare that we have read this thesis and in my
opinion this thesis is sufficient in term of scope and quality for the
award of the degree of Bachelor of Computer Science (Software Engineering)

Signature : _____
Name of Supervisor : DR. MUHAMMAD LUQMAN BIN MOHD
SHAFIE
Date : 17 APRIL 2025

DECLARATION

I declare that this thesis entitled “Al-Muneer Hall Reservations” is the result of my own research except as cited in the references. The thesis has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.

Signature :

Name : AHMED HANI GHALEB

Date : 17 APRIL 2025

DEDICATION

This thesis is dedicated to my father, who taught me that the best kind of knowledge to have is that which is learned for its own sake. It is also dedicated to my mother, who taught me that even the largest task can be accomplished if it is done one step at a time.

ACKNOWLEDGEMENT

In preparing this thesis, I was in contact with many people, researchers, academicians, and practitioners. They have contributed towards my understanding and thoughts. In particular, I wish to express my sincere appreciation to my main thesis supervisor, Professor Dr. Muhammad Luqman Bin Mohd Shafie, for encouragement, guidance, criticism and friendship. I am also very thankful to my co-supervisor Professor Dr Sim Hiew Moi and Associate Professor Dr. Shahida Sulaiman for their guidance, advice and motivation. Without their continued support and interest, this thesis would not have been the same as presented here.

My fellow students should also be recognised for their support. My sincere appreciation also extends to all my colleagues and others who have provided assistance on various occasions. Their views and tips are useful indeed. Unfortunately, it is not possible to list all of them in this limited space. I am grateful to all my family member.

ABSTRACT

Al-Muneer Hall for Weddings and Events, located in Ibb, Yemen, currently manages bookings and inquiries primarily through manual phone calls and direct messages. This traditional approach leads to significant time investment by the owner in addressing repetitive questions and managing processes manually, often overwhelming staff and potentially causing delays or inaccuracies for clients. Furthermore, potential customers, particularly female stakeholders adhering to local cultural norms who may prefer remote evaluation, lack a centralized, visually rich platform to explore the venue thoroughly and manage their interactions efficiently. The Al-Muneer Online Portal addresses this challenge by providing a dedicated, comprehensive web-based platform consolidating all essential information and key interaction processes into an accessible, user-friendly interface.

Leveraging modern web development practices, the portal will feature an interactive availability calendar, detailed pricing information, a comprehensive FAQ, and a high-quality visual gallery. Crucially, it will also incorporate functionalities for users to submit booking inquiries, optionally upload proof of payment (e.g., transfer screenshots) to confirm bookings, and provide feedback on their experience. For the administrator, the system will offer a secure panel to manage all content, availability, pricing, view submitted payment proofs, manage feedback, generate basic operational reports, and configure notifications (primarily via SMS/WhatsApp to align with local preferences). This approach aims to provide a centralized, reliable source of information and interaction, accessible 24/7, thereby significantly reducing the burden of manual inquiries and processes, enhancing the hall's marketing presence, streamlining operations, and improving the overall customer experience with cultural sensitivity.

ABSTRAK

Dewan Al-Muneer untuk Perkahwinan dan Acara, yang terletak di Ibb, Yaman, pada masa ini menguruskan tempahan dan pertanyaan terutamanya melalui panggilan telefon manual dan mesej langsung. Pendekatan tradisional ini menyebabkan pemilik perlu meluangkan masa yang signifikan untuk menjawab soalan berulang dan menguruskan proses secara manual, yang sering membebankan staf dan berpotensi menyebabkan kelewatan atau ketidaktepatan maklumat kepada pelanggan. Tambahan pula, bakal pelanggan, terutamanya pihak berkepentingan wanita yang mengikut norma budaya tempatan dan mungkin lebih suka membuat penilaian dari jauh, kekurangan platform berpusat yang kaya dengan visual untuk meneroka dewan secara menyeluruh dan menguruskan interaksi mereka dengan cekap. Portal Dalam Talian Al-Muneer menangani cabaran ini dengan menyediakan platform berasaskan web yang khusus dan komprehensif, menyatukan semua maklumat penting dan proses interaksi utama ke dalam antara muka yang mudah diakses dan mesra pengguna.

Dengan memanfaatkan amalan pembangunan web moden, portal ini akan menampilkan kalendar ketersediaan interaktif, maklumat harga terperinci, Soalan Lazim (FAQ) yang komprehensif, serta galeri visual berkualiti tinggi. Yang penting, ia juga akan menggabungkan fungsi untuk pengguna menghantar pertanyaan tempahan, secara pilihan memuat naik bukti pembayaran (cth., tangkapan skrin pindahan) untuk mengesahkan tempahan, dan memberikan maklum balas mengenai pengalaman mereka. Bagi pentadbir, sistem ini akan menawarkan panel selamat untuk mengurus semua kandungan, ketersediaan, harga, melihat bukti pembayaran yang dihantar, mengurus maklum balas, menjana laporan operasi asas, dan mengkonfigurasi pemberitahuan (terutamanya melalui SMS/WhatsApp untuk selaras dengan pilihan tempatan). Pendekatan ini bertujuan untuk menyediakan sumber maklumat dan interaksi berpusat yang boleh dipercayai, boleh diakses 24/7, sekali gus mengurangkan beban pertanyaan dan proses manual secara signifikan, meningkatkan kehadiran pemasaran dewan, melancarkan operasi, dan menambah baik pengalaman pelanggan secara keseluruhan dengan kepekaan budaya.

TABLE OF CONTENTS

TITLE	PAGE
CHAPTER 1 INTRODUCTION	1
1.1 Introduction	1
1.2 Problem Background	2
1.3 Project Aim	2
1.4 Project Objectives	3
1.5 Project Scope	4
1.6 Project Importance	5
1.7 Report Organization	6
CHAPTER 2 LITERATURE REVIEW	7
2.1 Introduction	7
2.2 Project Domain	7
2.3 Literature Review of Technology Used	15
2.4 Chapter Summary	17
CHAPTER 3 SYSTEM DEVELOPMENT METHODOLOGY	18
3.1 Introduction	18
3.2 Methodology Choice and Justification	18
3.3 Phases of the Chosen Methodology	21
3.4 Technology Used Description	23
3.5 Chapter Summary	25
CHAPTER 4 REQUIREMENT ANALYSIS AND DESIGN	26
4.1 Introduction	26
4.2 Requirement Analysis	26
4.2.1 Functional Requirements	27
4.2.2 Non-Functional Requirements	29
4.3 Project Design	30
4.3.1 System Architecture Diagram	30
4.3.2 Database Design	31
4.3.3 Use Case Table and Diagram	32
4.3.3.1 Selected Use Case: Submit Booking Inquiry (UC005)	32
4.3.3.2 Sequence Diagram (Submit Booking Inquiry - UC005)	34
4.3.3.3 Activity Diagram (Submit Booking Inquiry - UC005)	35
4.4 Interface Design (Low-Fidelity)	35
4.5 Chapter Summary	41
CHAPTER 5	43
CONCLUSION	43
5.1 Introduction	43
5.2 Achievement of Project Objectives	43
5.3 Future Improvements	44
References	46

Appendix A	49
Software Requirements Specification (SRS)	49
ChatUTM	49
Revision Page	49
a. Overview	49
b. Target Audience	49
c. Version Control History	49
1. Introduction	50
1.1 Purpose	50
1.2 Scope	50
1.3 Definitions, Acronyms, and Abbreviations	50
1.4 References	51
1.5 Overview	51
2. Specific Requirements	51
2.1 User Characteristics	51
2.1.1 UTM Students and Staff	51
2.2 System Features	52
Feature	52
Description	52
User Authentication	52
Users sign in via UTM SSO. Errors are displayed for invalid login attempts.	52
AI Chat Interface	52
Users can enter queries and receive AI-generated responses with references.	52
Suggested Questions	52
Frequently asked questions are provided for quick access.	52
Multilingual Support	52
The chatbot can process and respond in multiple languages.	52
Data Retrieval	52
Web scraping and RSS automation keep information up-to-date.	52
Provide Feedback	52
Users can rate responses as helpful or unhelpful.	52
Report Incorrect Information	52
Users can flag incorrect responses for review.	52
Chat Tab Navigation	52
Users can navigate through past conversations using a tab-based system.	52
Export Responses	52
Users can copy, generate a PDF, or create a public link for answers.	52
2.3 Use Case Details	52
Use Case Table	52
2.3.1 UC001: User Authentication	54
Sequence Diagram (User Authentication)	55
Activity Diagram (User Authentication)	55

2.3.2 UC002: Ask and Receive Answers	55
Sequence Diagram (Ask and Receive Answers)	56
Activity Diagram (Ask and Receive Answers)	57
2.3.3 UC003: View Suggested Questions	57
Sequence Diagram (View Suggested Questions)	58
Activity Diagram (View Suggested Questions)	58
2.3.4 UC004: Provide Feedback on Answers	59
Sequence Diagram (Provide Feedback on Answers)	60
Activity Diagram (Provide Feedback on Answers)	60
2.3.5 UC005: Report Incorrect Information	60
Sequence Diagram (Report Incorrect Information)	61
Activity Diagram (Report Incorrect Information)	61
2.3.6 UC006: Change Language Preference	62
Sequence Diagram (Change Language Preference)	62
Activity Diagram (Change Language Preference)	63
2.3.7 UC007: Navigation via Chat Tabs	63
Sequence Diagram (Navigation via Chat Tabs)	64
Activity Diagram (Navigation via Chat Tabs)	64
2.3.8 UC008: Export Answers	65
Sequence Diagram (Export Answers)	66
Activity Diagram (Export Answers)	66
2.4 Performance and Other Requirements	66
2.5 Design Constraints	67
Software Design Specification (SDS) for ChatUTM	68
Version 1.0	68
Prepared by: Ahmed Hani Ghaleb	68
Faculty of Computing, Universiti Teknologi Malaysia	68
1. Revision Page	68
1.1 Overview	68
1.2 Issuing Organization / Team	68
1.3 Authorship / Project Team Members	68
1.4 Revision History	69
2. Introduction	69
2.1 Purpose	69
2.2 Scope	69
2.3 Context	69
2.4 Summary	70
3. References	70
4. Glossary	70
5. Design Body	70
5.1 Design Stakeholders and Their Concerns	70
5.2 Context Viewpoint	71
5.2.1 Design Concerns	71
5.2.2 Design View (Use Case Diagram Placeholder)	71

5.3 Composition Viewpoint	72
5.3.1 Design Concerns	72
5.3.2 Design View (Architecture Diagram Placeholder)	72
5.4 Logical Viewpoint	72
5.4.1 Design Concerns	72
5.4.2 Design View (Class Diagram Placeholder)	72
5.5 Information Viewpoint	73
5.5.1 Design Concerns	73
5.6 Interface Viewpoint	74
5.6.1 Design Concerns	74
5.6.2 Design View (Interfaces Placeholder)	74
5.7 Structure Viewpoint	75
5.7.1 Design Concerns	75
5.7.2 Design View (Package Diagram Placeholder)	75
5.8 Interaction Viewpoint	76
5.8.1 Design Concerns	76
5.8.2 Design View (Sequence Diagrams Placeholder)	76
5.9 State Dynamic Viewpoint	77
5.9.1 Design Concerns	77
5.9.2 Design View (State Machine Diagram Placeholder)	78
5.10 Algorithm Viewpoint	78
5.10.1 Design Concerns	78
5.10.2 Design View (Activity Diagram Placeholder)	78
6. User Interface Design	79
6.1 Overview of User Interface	79
6.2 Screen Images (Placeholder for UI Screenshots)	79
7. Appendices	80
Software Test Documentation (STD) for ChatUTM	82
Version 1.0	82
Prepared by: Ahmed Ghaleb	82
Faculty of Computing, Universiti Teknologi Malaysia	82
1. Revision Page	82
1.1 Overview	82
1.2 Target Audience	82
1.3 Project Team Members	82
1.4 Version Control History	83
2. Introduction	83
2.1 Purpose	83
2.2 Scope	83
2.3 Definitions, Acronyms, and Abbreviations	84
2.4 References	84
2.5 System Overview	84
3. Test Cases, Data, and Expected Results	85
3.1 Functional Test Cases	85

Test TC001 for Module: User Authentication (UC001)	85
Test Case: TC001_01 – Login with Valid Credentials	85
Test Case: TC001_02 – Login with Invalid Credentials	86
3.2 Functional Test Cases for Query Processing (UC002)	87
3.3 Non-Functional Test Cases	87
4. Test Execution Instructions	88
4.1 Functional Testing	88
4.2 Performance Testing	88
4.3 Usability Testing	88
5. Test Results and Reporting	88
5.1 Test Log	88
5.2 Issue Tracking	88
5.3 Summary Report	88
6. Appendices	89
Appendix A: Traceability Matrix	89

LIST OF APPENDICES

APPENDIX	TITLE	PAGE
Appendix A	<u>Software Requirements Specification (SRS) for UTM Wikibot</u>	28
Appendix B	<u>Software Design Document (SDD) for UTM Wikibot</u>	47
Appendix C	<u>Software Test Document (STD) for UTM Wikibot</u>	52

CHAPTER 1

INTRODUCTION

1.1 Introduction

Al-Muneer Hall for Weddings and Events ("قاعة المنير للأفراح والمناسبات"), situated in Ibb, Yemen, is a venue primarily hosting weddings and other significant social gatherings, deeply embedded within the local cultural context. Currently, the hall's operations rely heavily on traditional communication methods, primarily phone calls and direct messaging, for managing customer inquiries, bookings, and follow-up confirmations. While functional, this manual process presents significant inefficiencies for both the business owner and potential clients seeking information or wishing to complete their booking process smoothly. This project proposes the development of the "Al-Muneer Online Portal," a dedicated web-based platform designed to modernize the hall's client interaction and operational management. The portal aims to serve as a comprehensive, accessible, and visually engaging information hub, integrating key functionalities such as inquiry submission, simple payment proof upload, feedback collection, owner-side reporting, and client notifications, thereby streamlining operations and enhancing the overall customer experience.

1.2 Problem Background

The current operational model of Al-Muneer Hall faces several challenges inherent in its reliance on manual communication and processes. The owner invests a disproportionate amount of time addressing repetitive inquiries regarding fundamental details such as venue availability, pricing, amenities, and visual representations. Beyond initial inquiries, managing booking confirmations (often involving verbal agreements or manual tracking of offline payments), collecting customer feedback systematically, and generating insights into business performance are also manual, time-consuming tasks. This constant influx of basic questions and administrative overhead consumes valuable time. Furthermore, this reliance on direct communication can lead to inconsistencies, limits accessibility for clients, and lacks a structured way to gather feedback for service improvement or provide timely updates and confirmations to clients.

A significant dimension to this problem lies within the socio-cultural context of Ibb, Yemen. Potential clients, particularly female stakeholders, lack a convenient way to evaluate the venue and manage interactions. The absence of a centralized online platform for information, simple booking confirmation steps (like uploading a payment transfer screenshot, common in a cash-predominant economy where formal banking is less prevalent), and structured feedback mechanisms creates barriers to informed decision-making and efficient service. This scattered and manual approach is inefficient, less professional, and does not leverage digital tools for effective marketing, customer service, or basic operational oversight.

1.3 Project Aim

The main aim of this project is to design, develop, and evaluate the Al-Muneer Online Portal, a web-based system intended to serve as the primary digital interface and operational support tool for Al-Muneer Hall for Weddings and Events. The goal is to centralize venue information, streamline the inquiry and

booking confirmation process, implement feedback collection and basic reporting for the owner, enable timely client notifications, reduce the owner's workload related to repetitive tasks, and provide an enhanced, accessible, and culturally considerate experience for potential clients.

1.4 Project Objectives

To achieve the project aim, the following objectives have been defined:

1. To elicit and analyze the requirements and needs of the Al-Muneer Hall owner and potential customers/visitors, focusing on features like an availability calendar, dynamic pricing display, FAQ section, media gallery, inquiry submission, simple payment proof upload, feedback mechanism, basic reporting dashboard, and notification preferences (SMS/WhatsApp).
2. To design the system architecture, database schema (ERD), and user interface/user experience (UI/UX) for the Al-Muneer Online Portal, incorporating modules for all core functionalities including content management, booking inquiries, payment proof handling, feedback management, reporting, and notifications, based on the gathered requirements.
3. To develop a functional web application featuring a visitor-facing front-end and an administrator back-end using Spring Boot (Java), implementing all specified features including secure admin controls for managing content, inquiries, payments, feedback, viewing reports, and triggering/managing notifications.
4. To evaluate and validate the developed portal through functional testing, usability testing, performance testing, and basic security testing, ensuring the reliability and effectiveness of all modules including the accuracy of reports,

the payment proof workflow, feedback submission and management, and notification delivery.

1.5 Project Scope

The scope of the Al-Muneer Online Portal project includes:

- **Platform:** A web-based application accessible via standard web browsers, featuring a responsive design for mobile-friendly viewing. No native mobile application.
- **Users:**
 - **Visitors:** Can browse venue information, view media gallery, check availability, view pricing, consult FAQ, submit booking inquiries, optionally upload payment proof (e.g., transfer screenshot), and submit feedback.
 - **Admin (Owner):** Can securely log in to manage hall information, media gallery, availability calendar, pricing, FAQ, view inquiries, view/manage submitted payment proofs (update status), manage feedback, view basic operational reports, and configure/manage notifications.
- **Core Functionalities:** Content display, interactive calendar, dynamic pricing display, FAQ, inquiry submission, simple payment proof upload and admin verification, feedback submission and admin review, basic reporting dashboard for admin, and admin-configurable notifications (primarily SMS/WhatsApp).
- **Limitations:**
 - Simple payment proof upload only (e.g., screenshot of local transfer receipt). No direct online payment processing or integration with banks/payment gateways.
 - No integration with external calendar systems.

- **No integrated live chat feature. Standard contact information (Hotline/WhatsApp number) will be clearly displayed.**
- No advanced event management or Customer Relationship Management (CRM) features.
- Analytics will be basic (e.g., visitor counts, inquiry logs contributing to reports).
- Owner provides initial content.
- Website anticipates moderate traffic.
- **Reporting will be basic** (e.g., summaries of inquiries, bookings based on payment status, feedback overview).
- **Notifications primarily via SMS/WhatsApp reflecting local preferences;** email notifications will be minimal or optional.

1.6 Project Importance

The development of the Al-Muneer Online Portal holds significant importance. Firstly, it addresses operational inefficiencies by automating information dissemination and streamlining processes like booking confirmation and feedback collection. Secondly, it enhances the customer's experience by providing 24/7 access to comprehensive information and clear next steps for inquiry and booking.

Crucially, within the Yemeni cultural context, the portal offers a vital tool for inclusivity and efficient interaction. It allows remote evaluation and facilitates simpler booking confirmation through locally understood payment proof methods. The inclusion of feedback mechanisms allows the owner to continuously improve services based on client input. Basic reporting provides the owner with valuable insights into operational trends without complex tools. Timely notifications improve communication and client satisfaction. This portal represents a necessary step in digitizing business operations to meet contemporary expectations and improve overall service delivery.

1.7 Report Organization

This report is structured into five chapters. Chapter 1 provides an introduction to the project, outlining the problem background, project aim, objectives, scope, and its importance. Chapter 2 presents a literature review covering the relevant domain of online reservation systems, event venue management platforms, including features like payment handling, feedback systems, reporting, and notification mechanisms, user interface design principles, and the technologies planned for use. Chapter 3 will detail the methodology adopted for the system development lifecycle. Chapter 4 will discuss the system analysis and design, including requirements specification, architectural design, database design, and UI/UX design. Finally, Chapter 5 will conclude the report, summarizing the project work, discussing the findings, and suggesting potential areas for future enhancement.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter explores the existing literature and technological foundations relevant to the development of the Al-Muneer Online Portal. It delves into the domain of online event venue management systems, examining the shift from traditional manual operations to digital platforms that offer enhanced functionality. The discussion covers key features expected in modern venue portals, including mechanisms for inquiry management, simple booking confirmation processes (such as payment proof submission), feedback collection, basic reporting dashboards, and client notification systems. This chapter also presents a case study of Al-Muneer Hall to provide context for the system's development. It analyzes the limitations of current manual processes, and compares the proposed system with existing solutions or alternative approaches. Furthermore, this chapter reviews the literature concerning the specific technologies selected for the portal's development, justifying their suitability for implementing these comprehensive features (O'Brien & Marakas, 2011).

2.2 Project Domain

The Al-Muneer Online Portal falls within the domain of specialized web-based information, inquiry management, and operational support systems for the event management and hospitality industry. Such systems aim to bridge the communication gap between venue providers and potential clients, offering efficiency, accessibility, enhanced transactional capabilities, and improved

post-interaction engagement compared to traditional methods (Sigala et al., 2012; Buhalis & Law, 2008).

2.2.1 Key Features and Information Architecture for Event Venue Portals

Modern event venue websites have evolved beyond simple static brochures to become interactive platforms. Literature highlights several critical components for success:

- **Visual Appeal and Information Richness:** High-quality imagery and detailed descriptions remain crucial (Ukpabi & Karjaluoto, 2017).
- **Availability Information:** Interactive calendars are a core expectation for instant information (Werthner & Ricci, 2004).
- **Pricing Transparency:** Clear pricing, even if base rates, is valued (Hayes & Miller, 2011).
- **Clear Call-to-Actions and Inquiry Forms:** Essential for lead generation (Nielsen, 2000).
- **Simple Booking Confirmation/Payment Proof:** In contexts where full online payment gateways are not feasible or common, providing a mechanism for users to submit proof of offline payment (e.g., bank transfer receipt, payment app screenshot) is a practical alternative to streamline booking confirmation. This reflects an adaptation to local payment ecosystems (Kshetri, 2013, on mobile payments in developing economies).
- **Feedback Mechanisms:** Incorporating systems for users to provide feedback on the venue or service is vital for continuous improvement and demonstrating customer centricity. Online feedback forms are a common implementation (Morgan et al., 2003).
- **Client Notification Systems:** Automated or semi-automated notifications (e.g., for inquiry received, booking confirmed, payment acknowledged) enhance customer communication and reduce uncertainty. The choice of channel (email, SMS, WhatsApp) should align with user preferences and

local infrastructure (Kaplan & Haenlein, 2010, on social media and mobile communication).

- **Basic Reporting for Administrators:** For venue owners, access to simple reports on inquiries, booking statuses, and feedback trends provides valuable insights for operational management and decision-making without requiring complex business intelligence tools (Davenport & Harris, 2007).
- **Accessibility and Usability:** The portal must be easily navigable, responsive, and culturally appropriate (W3C, 2018; Marcus & Gould, 2000).

The information architecture should logically integrate these functionalities, ensuring a seamless user journey from initial exploration to inquiry, potential payment proof submission, and feedback.

2.2.2 Manual Operation Analysis

The current manual system at Al-Muneer Hall is common among SMEs but suffers from inefficiencies:

- **Time Consumption:** Staff spend time on repetitive queries and manual tracking of bookings, payments, and feedback. (Laudon & Laudon, 2016).
- **Limited Availability:** Information and process progression are tied to staff hours.
- **Inconsistency:** Information can vary; tracking booking confirmations and feedback lacks standardization.
- **Lack of Scalability:** Handling increased volume becomes difficult.
- **Poor User Experience:** Customers expect streamlined online interactions, including clear confirmation steps and ways to voice opinions (Sigala et al., 2012).
- **Documentation and Oversight Challenges:** Manual tracking makes it hard to generate operational insights or reliably follow up on client interactions.

Transitioning to a dedicated portal addresses these by automating information, standardizing processes for booking confirmation and feedback,

providing 24/7 accessibility, and offering basic reporting for better oversight (Chaffey & Ellis-Chadwick, 2019).

2.2.3 Current System Analysis - Comparison of Existing Systems/Approaches

Before presenting a direct comparison, it is pertinent to describe the categories of existing systems or approaches that businesses like Al-Muneer Hall might currently use or consider. The current manual process, as employed at Al-Muneer Hall and prevalent in many small to medium-sized enterprises (SMEs), relies heavily on direct human interaction for all customer engagement and booking management. Communication predominantly occurs via phone calls, face-to-face meetings, and informal messaging applications such as WhatsApp. While this approach offers high personalization, it is labor-intensive, possesses limited scalability, and risks inconsistencies in information delivery and record-keeping (Harrigan et al., 2012). Bookings, payment confirmations, and feedback are typically handled in an ad-hoc manner, lacking a structured or centralized system.

Another category includes generic online booking platforms, such as Eventbrite or Cvent. These third-party services provide a broad array of features for event listing, ticketing, and sometimes comprehensive event management. They usually offer a standardized interface for users to discover and book events or services. Despite their power, they might impose their own branding, fee structures, and workflows that may not align with the unique needs or cultural context of an independent venue (Gazzaroli et al., 2019). Their primary focus is often on ticketed events, which differs from bespoke venue rental inquiries.

Furthermore, simple website builders like Wix, Squarespace, or GoDaddy Website Builder enable individuals and small businesses to create websites with relative ease using pre-designed templates and drag-and-drop interfaces (Odoom et al., 2017). These platforms are excellent for establishing a basic online presence and displaying information. However, their inherent functionalities for dynamic processes such as custom availability calendars, managed pricing specific to complex services, integrated payment proof workflows, or backend reporting are often limited unless augmented by third-party plugins, which can introduce additional complexity or costs.

Finally, in many local business environments, it is common to encounter typical static competitor websites. These sites generally function as online brochures, providing contact information, a gallery, and perhaps a list of services. They typically lack interactive features for checking availability, obtaining dynamic pricing, submitting structured inquiries, or managing other transactional aspects online. Their primary role is informational, still necessitating users to make contact via phone or email for most substantive interactions.

A comparative analysis helps to position the proposed Al-Muneer Online Portal:

Feature	Al-Muneer Online Portal (Proposed)	Manual (Phone/Whats App)	Generic Booking Platforms [Gazzaroli et al., 2019]	Simple Website Builders [Odoom et al., 2017]	Static Competitor Website
Venue-Specific Info	Tailored	(Manual)	Generic Structure	(Manual Setup)	Basic
Interactive Calendar	Yes	No	Often (Complex Setup)	Limited/Plugging Dependent	Often No
Dynamic Pricing Display	Yes (Admin Managed)	(Verbal/Manual)	Sometimes (Complex)	Usually No	Often No
Admin Panel	Custom Built	N/A	Yes (Platform Specific)	(General CMS)	N/A
Payment Proof Upload	Yes (Simple)	(Manual Exchange)	Often Not Direct	No	No
Feedback System	Yes	(Ad-hoc)	Sometimes (Platform)	Usually No	No

Basic Reporting	Yes	No	Limited (Platform)	No	No
Notifications (SMS/WA)	Yes (Focused)	(Manual)	Email Often Primary	No	No
Cultural Context Adapt	High (Custom Design)	High (Direct Interaction)	Low (Standardized)	Medium (Template Dependent)	Medium

Analysis of Existing Systems/Approaches:

The manual process is highly inefficient and lacks structured mechanisms for confirmation, feedback, or oversight. Generic booking platforms may offer some advanced features but often lack the tailored simplicity needed (especially for payment processes in specific economic contexts) and may not align with local communication preferences like WhatsApp/SMS for notifications (Buhalis & Law, 2008). Simple website builders generally do not offer integrated backend functionalities for managing payment proofs, feedback systematically, or generating specific operational reports. The proposed Al-Muneer Online Portal is designed to provide a holistic, tailored solution that addresses the venue's unique operational flow, from inquiry to simple booking confirmation and feedback, all managed through a custom-built admin panel offering relevant insights and communication tools suited to the local environment (Laudon & Laudon, 2016).

2.2.4 Case Study: Al-Muneer Hall for Weddings and Events

To ground the project in a real-world context, this section details Al-Muneer Hall, the business for which the proposed system is being developed. Al-Muneer Hall ("قاعة المنير للأفراح والمناسبات") is an event venue located in Ibb, Yemen. Its primary business is hosting weddings, catering to the local community and upholding traditional Yemeni cultural practices associated with such events. The hall also accommodates other social gatherings, such as consolation meetings and special occasions, offering a versatile space for various functions. The business operates

from a single physical location. In terms of its structure, Al-Muneer Hall is an owner-operated business. The primary stakeholder is Mr. Ahmed Almunajid, who is actively involved in the day-to-day management and client interactions. While there may be ancillary staff for event setup, cleaning, and support during events, the core administrative, booking, and client communication tasks are predominantly handled or directly overseen by Mr. Almunajid himself or his manager. This centralized management structure underscores the need for tools that can enhance efficiency and reduce the owner's direct operational burden.

The primary method of engagement for this project involved direct interaction with the owner, Mr. Ahmed Almunajid. Initial discussions were conducted via a scheduled video conference (Google Meet) on April 4, 2025, to understand the business, its current processes, challenges, and his vision for an online portal. This was followed by ongoing asynchronous communication via WhatsApp for clarifications and to gather specific details, as evidenced by communication logs and the NABC proposal document. Key findings from this engagement, effectively forming the basis of a user interview and requirements gathering phase, revealed a significant portion of Mr. Almunajid's time is spent answering repetitive phone calls regarding availability, pricing, venue capacity, and visual details. There was a clear desire for a professional online presence to accurately showcase the hall. Specific feature requests included an interactive calendar, a clear way to display pricing, a media gallery, and an FAQ section. The current booking process involves phone or in-person arrangements, with payment typically in cash, although local bank transfers with screenshot proof are emerging. There is no formal system for tracking these proofs or managing feedback. Communication preferences lean heavily towards WhatsApp and direct calls over email, reflecting local business practices. A core need identified was for a system that is easy for Mr. Almunajid to manage and for his clients to use. Following discussions about the initial proposal feedback, Mr. Almunajid expressed openness to incorporating a simple mechanism for clients to upload payment proof, a feature for clients to submit feedback, basic reports on inquiries and booking statuses, and automated notifications, preferably via SMS or WhatsApp. These interactions confirmed the problem statement and helped refine the scope and objectives of the Al-Muneer Online Portal project, ensuring its alignment with the stakeholder's genuine needs.

2.3 Literature Review of Technology Used

The selection of technologies for the Al-Muneer Online Portal is based on their suitability for developing a robust, scalable, and maintainable web application that meets the defined requirements, including the newly added functionalities. For each core technology, alternatives were considered.

Technology	Purpose	Alternatives Considered	Why Chosen / Developer Justification
Spring Boot (Java) / MVC	Backend framework	Python (Django/Flask), Node.js (Express.js), PHP (Laravel)	I'm already experienced in Spring MVC but that is not the main reason, Spring boot gives you everything you need from authentication to the design pattern infrastructure so you only focus on implementing the idea because the heavy lifting has been done for us in a complete nicely wrapped backend framework.
MySQL / PostgreSQL	Relational database	NoSQL (MongoDB, Cassandra), SQLite, MS SQL Server	There is no need for NoSQL database for this project because we're not dealing with unstructured type of data. PostgreSQL sounds a good solid choice. Everyone loves it. It does the job perfectly and it's open source.
HTML5, CSS3, JS (Responsive)	Frontend development	JavaScript Frameworks (React, Angular, Vue.js)	"An idiot admires complexity, a genius admires simplicity" — Terry Davis, Creator of Temple OS
Waterfall Model	SDLC methodology	Agile (Scrum, Kanban), Spiral Model	Sounds solid when having clear requirements of a moderate software that does not seem to be

			needing frequent and high-maintenance in post development
Cloud Hosting (VPS)	Deployment environment	AWS, serverless	VPS is the safest option against DDoS attacks that could cause massive cloud computing bills
Admin Auth (Hashed Pass.)	Secure admin login	Plain text storage (highly insecure), Reversible encryption	The encryption standard in web development
Notification Mechanisms	(e.g., SMS Gateway API, Email libraries, Manual WhatsApp)	In-app notifications (not a native app), Full WhatsApp Business API integration	notifications sound simple to implement though it's very effective compared to the other complex means that the stakeholder does not really need given the complexity and return on that.

The Model-View-Controller (MVC) pattern, facilitated by Spring Boot, ensures separation of concerns, making the application easier to manage and extend (Fowler, 2002). Relational databases like MySQL or PostgreSQL provide data integrity crucial for managing bookings and venue information (Garcia-Molina et al., 2008). Responsive design using HTML5, CSS3, and JavaScript is non-negotiable for modern web applications to cater to diverse user devices (W3C, 2018). While Agile methods are popular, the Waterfall model's sequential nature can provide structure for a project with initially clear requirements, as outlined in the proposal phase (Royce, 1970). Security via HTTPS and proper password hashing is paramount for user trust and data protection (Rescorla, 2001; Schneier, 1996). Cloud hosting offers flexibility and reliability for deployment (Armbrust et al., 2010).

2.4 Chapter Summary

This chapter reviewed the literature pertinent to the Al-Muneer Online Portal project, incorporating insights from the newly added case study of Al-Muneer Hall which detailed stakeholder engagement and specific business needs. It established the context within online event venue management systems, highlighting the necessity of core informational features alongside functionalities for simple booking confirmation, systematic feedback collection, basic operational reporting, and client notifications. The limitations of manual processes in these extended areas were emphasized. The discussion of existing systems was expanded to describe each category before comparison. A comparative analysis reinforced that the custom portal offers a uniquely tailored solution. Finally, the chapter justified the selection of the technology stack, including a comparison with alternatives for each key component, confirming their suitability for achieving the project's comprehensive objectives.

CHAPTER 3

SYSTEM DEVELOPMENT METHODOLOGY

3.1 Introduction

Developing the Al-Muneer Online Portal, a system with well-defined requirements and a clear scope, necessitates a structured and systematic approach to ensure all objectives are met efficiently and effectively. The chosen methodology must guide the project through its lifecycle, from initial concept through to the design and planning stages relevant to this PSM1 phase, and subsequently through development and deployment in PSM2. This chapter details the system development methodology selected for the Al-Muneer Online Portal, outlines its various phases, and describes how it will be applied to achieve the project's goals. The Waterfall model has been selected as the guiding methodology, providing a sequential and disciplined process well-suited for a project of this nature, where requirements are understood upfront and a step-by-step progression is preferred.

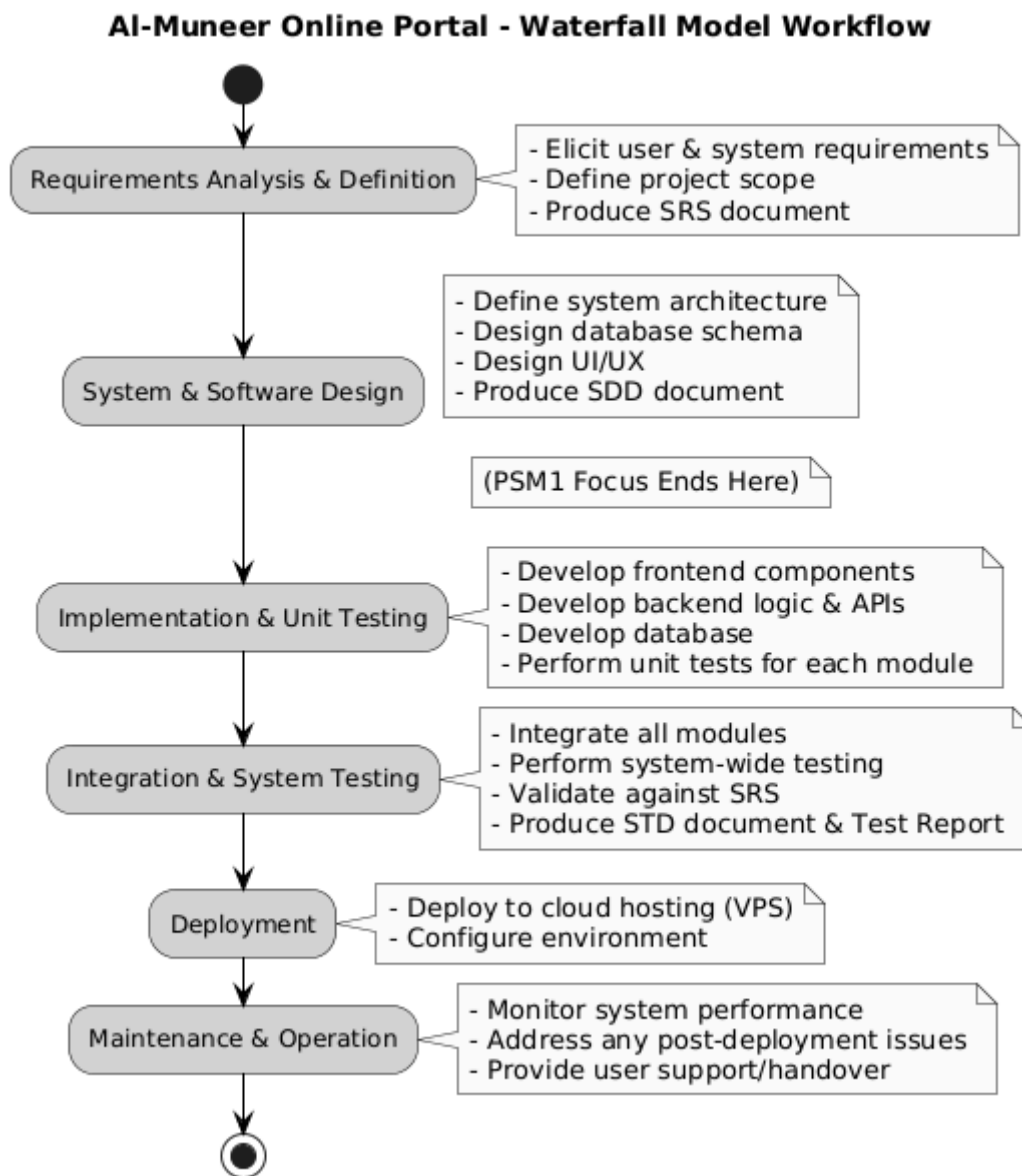
3.2 Methodology Choice and Justification

The Waterfall model was selected for the Al-Muneer Online Portal project due to its linear, sequential approach to software development. This model divides the project lifecycle into distinct phases, where each phase must be fully completed before the next begins. This structured nature is particularly advantageous for projects where requirements are well-understood and documented early on, as is the case with the Al-Muneer Online Portal following initial stakeholder engagement and

proposal refinement. The clarity of scope for features such as the interactive calendar, pricing display, payment proof upload, feedback system, basic reporting, and notifications makes the Waterfall model a suitable choice.

Traditional development models like Waterfall are often favored for projects where predictability and thorough documentation are key. For a student-led Final Year Project (FYP) with defined academic milestones (Progress 1, Progress 2, Final Report), the distinct phases of Waterfall provide clear checkpoints for progress assessment and deliverables, such as the Software Requirements Specification (SRS) and System Design Document (SDD). While Agile methodologies offer flexibility for projects with evolving requirements, the Al-Muneer Online Portal benefits more from the Waterfall model's emphasis on upfront planning and design, minimizing ambiguity as the project transitions from the planning and design stages in PSM1 to development in PSM2. This approach ensures that development work in PSM2 can proceed based on a stable and agreed-upon foundation, which is crucial for managing the project scope within the academic timeframe. The sequential nature also simplifies project management for a solo developer.

3.2.1 Waterfall Model Process Diagram



3.3 Phases of the Chosen Methodology

The development lifecycle for the Al-Muneer Online Portal, following the Waterfall model, is composed of distinct phases. Each phase has specific objectives and deliverables, ensuring a systematic progression towards the final product.

1. Requirements Analysis and Definition

This initial phase is critical for understanding and documenting the precise needs of the Al-Muneer Online Portal. It involves detailed discussions with the stakeholder (Mr. Almunajid) to elicit all functional and non-functional requirements. Activities include identifying user roles (Visitor, Admin), defining features such as venue information display, media gallery, interactive availability calendar, pricing customization panel, booking inquiry submission, simple payment proof upload, feedback submission, admin management of content, payment status updates, feedback review, basic report generation, and notification configuration. The primary deliverable of this phase is the Software Requirements Specification (SRS) document, which serves as a foundational agreement on what the system will do. This phase was a significant part of the early PSM1 activities.

2. System and Software Design

Once the requirements are clearly defined and documented in the SRS, the design phase begins. In this phase, the overall system architecture for the Al-Muneer Online Portal is established. This includes designing the database schema (for MySQL/PostgreSQL), detailing the structure of the backend (Spring Boot services and APIs), and designing the user interface (UI) and user experience (UX) for both the visitor-facing front-end and the administrator panel. Decisions about module breakdown, data flow, and security considerations are made here. The key deliverable for this phase is the System Design Document (SDD), which provides a blueprint for the development team (the student developer). This phase is a core component of PSM1, particularly leading up to Progress 2.

3. Implementation and Unit Testing

This phase marks the commencement of actual coding and development, which will primarily occur during PSM2. Based on the SDD, the Al-Muneer

Online Portal will be built. The backend logic using Spring Boot (Java) and the frontend using HTML, CSS, and JavaScript will be developed. The database will be created and populated with initial structures. As each module or component is developed, it will undergo unit testing to verify that it functions correctly in isolation according to its design specifications.

4. Integration and System Testing

After individual units have been developed and tested, they are integrated into a complete system. This integrated system then undergoes rigorous system testing to ensure that all components work together harmoniously and that the system as a whole meets the requirements specified in the SRS. This includes functional testing, usability testing, performance testing (basic), and security testing. A Software Test Document (STD) will guide this process, and a test report will summarize the findings. This phase is also planned for PSM2.

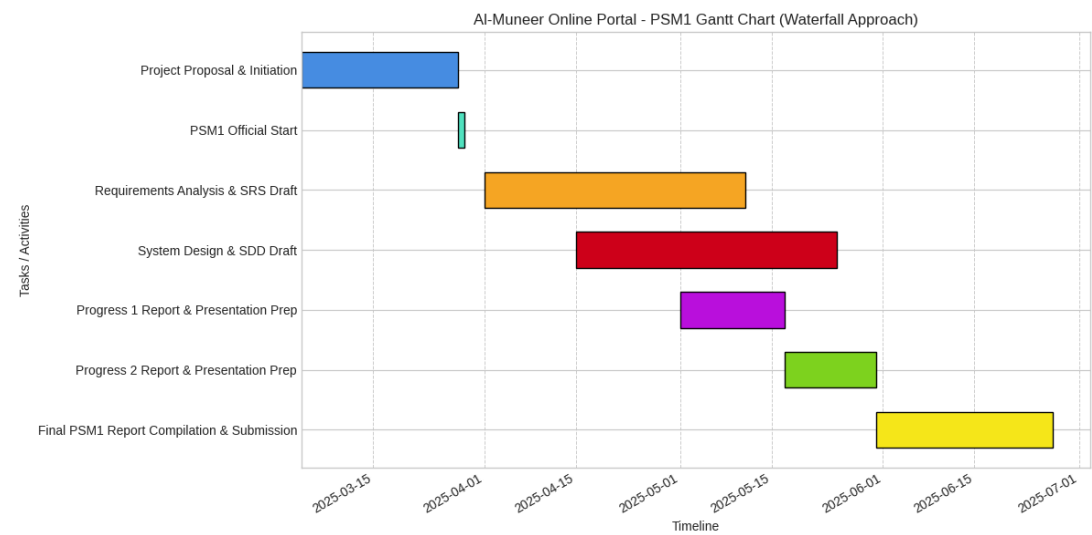
5. Deployment

Upon successful completion of system testing, where the Al-Muneer Online Portal is deemed stable and meets all requirements, it will be deployed to a production-like environment. For this project, deployment to a Cloud VPS is planned. This phase involves setting up the hosting environment, deploying the application code and database, and performing final checks to ensure the portal is live and accessible to users. This will occur towards the end of PSM2.

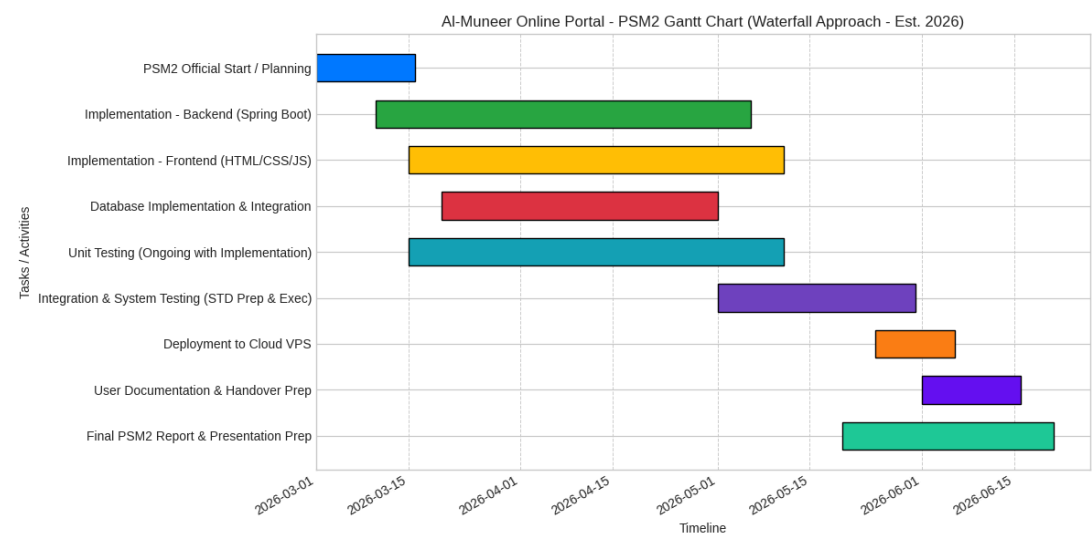
6. Maintenance and Operation

Following deployment, the maintenance phase begins. This involves ongoing monitoring of the Al-Muneer Online Portal for any issues, bugs, or required updates. For the scope of the FYP, this phase might include a period of observation, addressing any immediate post-deployment fixes, and preparing documentation for the handover of the system to the stakeholder. While extensive long-term maintenance is beyond the scope of an FYP, ensuring the system is stable and functional upon delivery is crucial.

3.3.1 Gantt Chart for PSM1



3.3.2 Gantt Chart for PSM2



3.4 Technology Used Description

The development of the Al-Muneer Online Portal, following the Waterfall methodology, integrates several key technologies systematically across its phases.

During the System and Software Design phase (PSM1), the architecture defines how these technologies interact. The backend, built using Spring Boot (Java) under an MVC pattern, will house the core business logic. This includes modules for managing venue information, handling booking inquiries, processing payment proofs, managing feedback, generating reports, and triggering notifications. Spring Security will be integral for protecting the admin panel.

The MySQL/PostgreSQL relational database, designed during this phase, will store all persistent data, including user credentials (hashed), hall details, availability, pricing tiers, inquiry records, payment proof metadata, and feedback submissions. Its structured nature will directly support the data retrieval needed for the admin's reporting features.

For the frontend, HTML5, CSS3, and JavaScript will be used to create the responsive user interface for both visitors and the administrator. This includes forms for inquiries, payment proof uploads, and feedback, as well as displays for the calendar, gallery, and dynamic pricing. These will be designed in PSM1 and implemented in PSM2.

Notification mechanisms, designed with a preference for SMS/WhatsApp channels, will involve integrating with an SMS gateway API where feasible during the implementation phase (PSM2), or by designing admin-triggered alerts. Email libraries within Spring Boot may support minimal email confirmations if required.

The deployment phase will utilize a Cloud VPS, where the packaged Spring Boot application and the database will be hosted. Secure communication via HTTPS (SSL/TLS) will be enforced to protect all data exchanged between the client browsers and the server. The Waterfall model ensures that each of these technology components is designed, then implemented, and finally tested in a sequential and organized manner.

3.5 Chapter Summary

This chapter outlined the System Development Methodology chosen for the Al-Muneer Online Portal, which is the Waterfall model. The rationale for selecting this structured, sequential approach was discussed, highlighting its suitability for a project with well-defined requirements and clear academic milestones. The distinct phases of the Waterfall model—Requirements Analysis and Definition, System and Software Design, Implementation and Unit Testing, Integration and System Testing, Deployment, and Maintenance—were detailed in the context of the project's progression through PSM1 (focusing on analysis and design) and into PSM2 (focusing on implementation, testing, and deployment). Illustrative Gantt charts for both PSM1 and PSM2 were provided to visualize the project timeline and key activities within these phases. Finally, a description was provided of how the selected technologies will be integrated and utilized throughout the development lifecycle as guided by the Waterfall methodology, ensuring a systematic approach to building the Al-Muneer Online Portal.

CHAPTER 4

REQUIREMENT ANALYSIS AND DESIGN

4.1 Introduction

This chapter outlines the functional and non-functional requirements for the Al-Muneer Online Portal. It further details the system architecture, database structure, user interface design concepts, and key process flows. The primary goal of this phase, executed within PSM1, is to translate the stakeholder needs and project objectives identified in earlier phases into a comprehensive blueprint for the system's development in PSM2. The design centers on providing an intuitive and efficient experience for both prospective clients (Visitors) inquiring about Al-Muneer Hall and the administrator (Owner) managing the portal's content and operations. The requirements and design elements documented herein will form the basis for the Software Requirements Specification (SRS) and System Design Document (SDD).

4.2 Requirement Analysis

Requirement analysis is the process of defining the expectations of the users and the attributes of the system. For the Al-Muneer Online Portal, requirements were gathered primarily through direct interaction with the stakeholder (Mr. Almunajid, owner of Al-Muneer Hall), as detailed in the Case Study (Chapter 2.2.4), and an analysis of common functionalities expected from similar online venue management systems, tailored to the local context of Yemen.

4.2.1 Functional Requirements

Functional requirements describe what the system should do. For the Al-Muneer Online Portal, these are categorized by user roles and key system operations:

FR1: Visitor Functionalities

- **FR1.1 View Venue Information:** The system shall display comprehensive information about Al-Muneer Hall, including descriptions, capacity, services offered, and location details.
- **FR1.2 View Media Gallery:** The system shall allow visitors to browse a gallery of images and potentially videos showcasing the hall.
- **FR1.3 Check Availability:** The system shall provide an interactive calendar allowing visitors to view dates that are already booked and those that are available.
- **FR1.4 View Pricing Information:** The system shall display pricing information. This may include base rates and options for customizable packages or services that affect the price.
- **FR1.5 View FAQ:** The system shall display a list of frequently asked questions and their answers.
- **FR1.6 Submit Booking Inquiry:** The system shall provide a form for visitors to submit a booking inquiry, including their name, contact details, desired date(s), and any specific requests.
- **FR1.7 Submit Payment Proof (Optional):** The system shall allow visitors, after submitting an inquiry or upon instruction from the admin, to optionally upload an image file (e.g., screenshot of a local transfer receipt) as proof of payment. The system shall only accept common image formats (e.g., JPG, PNG) and enforce a reasonable file size limit.
- **FR1.8 Submit Feedback:** The system shall provide a form for visitors to submit feedback about the venue or their experience with the portal.

FR2: Administrator Functionalities

- **FR2.1 Secure Admin Login:** The system shall require administrators to log in using unique, authenticated credentials (username and hashed password). The system shall display an error message if authentication fails.
- **FR2.2 Manage Venue Information:** The system shall allow the administrator to create, update, and delete textual information about the hall (descriptions, services, FAQs).
- **FR2.3 Manage Media Gallery:** The system shall allow the administrator to upload new images/videos to the gallery and delete existing ones.
- **FR2.4 Manage Availability Calendar:** The system shall allow the administrator to mark dates as booked or available on the interactive calendar.
- **FR2.5 Manage Pricing:** The system shall allow the administrator to set and update base pricing and any customizable pricing options.
- **FR2.6 Manage Booking Inquiries:** The system shall allow the administrator to view submitted booking inquiries.
- **FR2.7 Manage Payment Proofs:** The system shall allow the administrator to view uploaded payment proofs and update the status of a booking (e.g., "Payment Verified," "Pending Confirmation").
- **FR2.8 Manage Feedback:** The system shall allow the administrator to view submitted user feedback.
- **FR2.9 Generate Basic Reports:** The system shall allow the administrator to view basic reports, such as a list of inquiries within a date range, a summary of booking statuses, or an overview of feedback received.
- **FR2.10 Manage Notifications:** The system shall allow the administrator to configure triggers or templates for notifications (e.g., inquiry received, payment proof acknowledged) to be sent via preferred channels (SMS/WhatsApp, with email as a secondary option).

FR3: General System Functionalities

- **FR3.1 Responsive Design:** The system shall be responsive and accessible on common web browsers across various devices (desktops, tablets, smartphones).

- **FR3.2 Secure Data Handling:** The system shall securely handle all user-submitted data, including personal information in inquiries, uploaded payment proofs, and feedback.
- **FR3.3 Notification Generation:** Based on admin configuration and specific system events (e.g., new inquiry, payment proof submission), the system shall generate notifications to be sent to relevant parties (client and/or admin).

4.2.2 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system.

- **NFR1 Performance:** The system should load pages within an acceptable time frame (e.g., 3-5 seconds on a typical internet connection). The interactive calendar should respond promptly to user interactions.
- **NFR2 Usability:** The portal should be intuitive and easy to navigate for both visitors and the administrator. The language used will be primarily Arabic, with an option for English if deemed feasible within the project scope. UI elements should be clear and user-friendly, considering local cultural norms for design.
- **NFR3 Security:**
 - **NFR3.1 Admin Authentication:** The admin login shall be protected against common vulnerabilities. Passwords must be hashed.
 - **NFR3.2 Data Protection:** User-submitted data and uploaded files (payment proofs) must be protected from unauthorized access.
 - **NFR3.3 SSL/TLS:** All communication between the client and server shall be encrypted using HTTPS.
- **NFR4 Reliability:** The system should be available for use with minimal downtime. Database backups should be considered as part of operational reliability.
- **NFR5 Maintainability:** The code should be well-structured, commented, and follow the chosen MVC pattern to facilitate future updates or modifications (relevant for PSM2 and beyond).

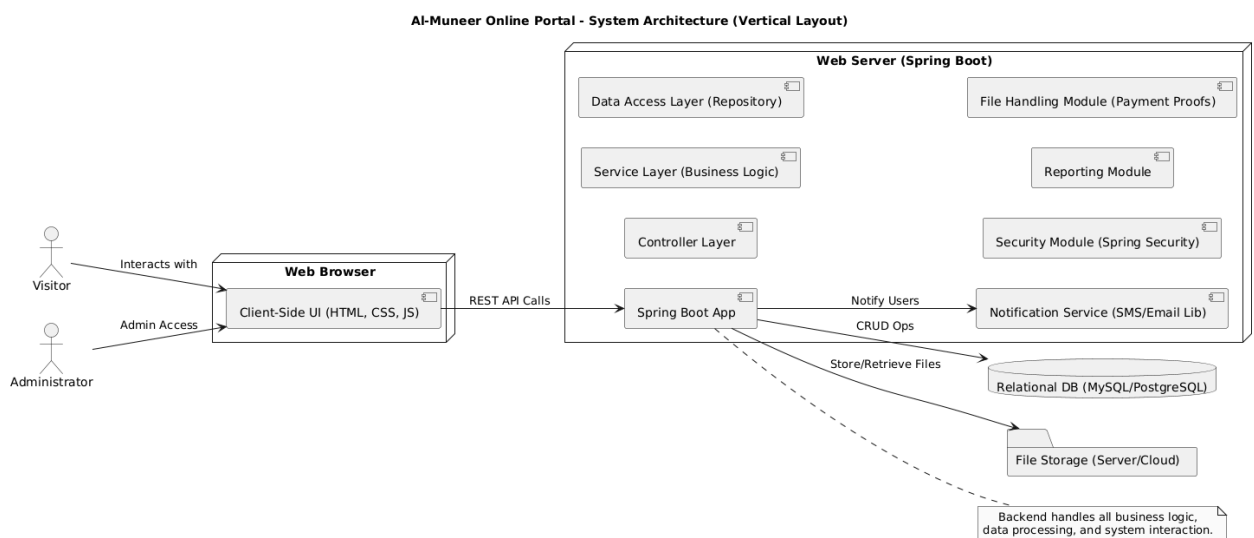
- **NFR6 Scalability (Basic):** While not expecting massive concurrent users initially, the system architecture (Spring Boot, chosen database) should be capable of handling a moderate increase in traffic (up to 10,000 monthly visitors as per initial scope) without significant performance degradation.

4.3 Project Design

The Al-Muneer Online Portal employs a layered architectural design to ensure separation of concerns, maintainability, and scalability. This design consists of a presentation layer (frontend), an application layer (backend logic), and a data layer (database).

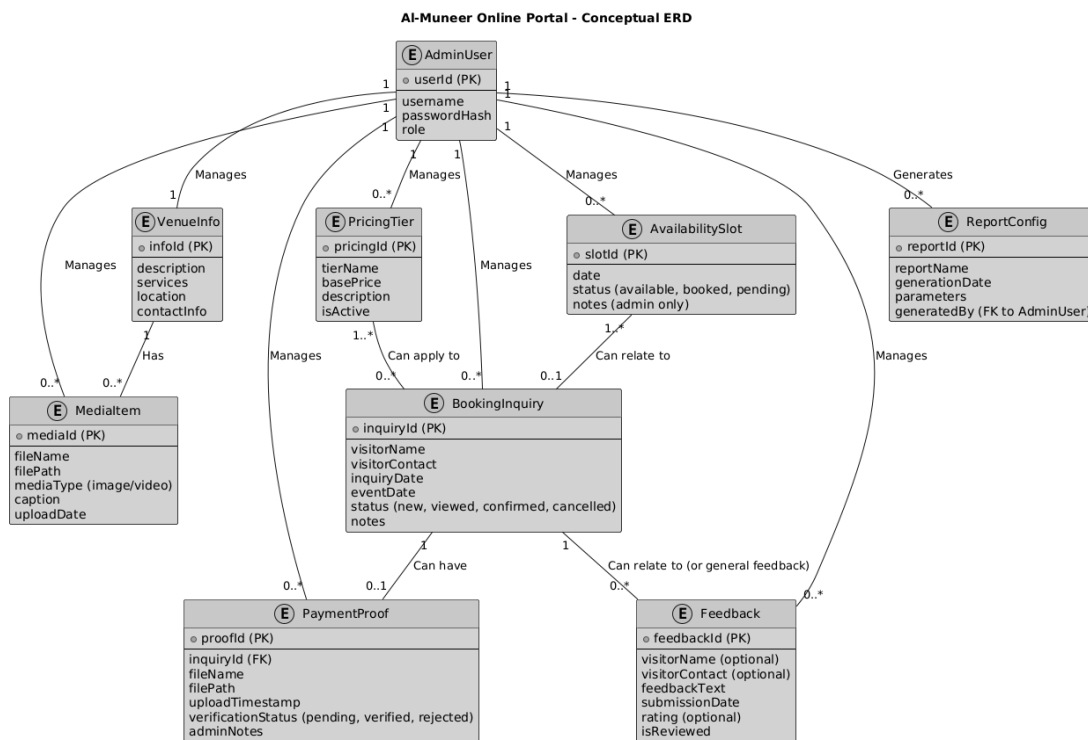
4.3.1 System Architecture Diagram

The following diagram illustrates the high-level architecture of the Al-Muneer Online Portal:



4.3.2 Database Design

A relational database model is chosen due to the structured nature of the data. The following is a simplified conceptual Entity-Relationship Diagram (ERD). Primary Keys (PK) and Foreign Keys (FK) would be defined in a detailed design.

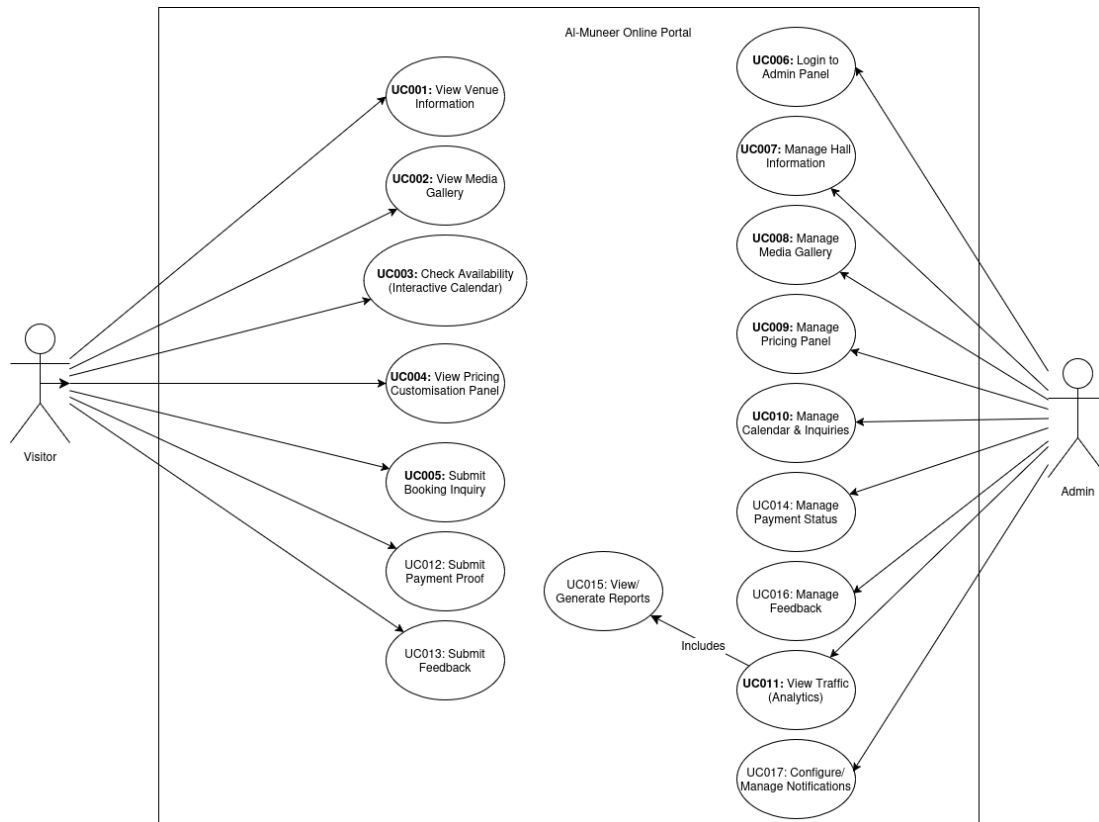


Data Components:

- **AdminUser:** Stores admin credentials and roles.
- **VenueInfo:** General descriptive information about the hall.
- **MediaItem:** Details about uploaded images and videos for the gallery.
- **AvailabilitySlot:** Represents specific dates and their booking status.
- **PricingTier:** Information about different pricing packages or rates.
- **BookingInquiry:** Details of inquiries submitted by visitors.
- **PaymentProof:** Metadata for uploaded payment proof files, linked to inquiries.
- **Feedback:** Stores feedback submitted by users.

- **ReportConfig:** (Conceptual) Could store metadata about generated reports if needed for history, though simple reports might be generated on-the-fly.

4.3.3 Use Case Table and Diagram

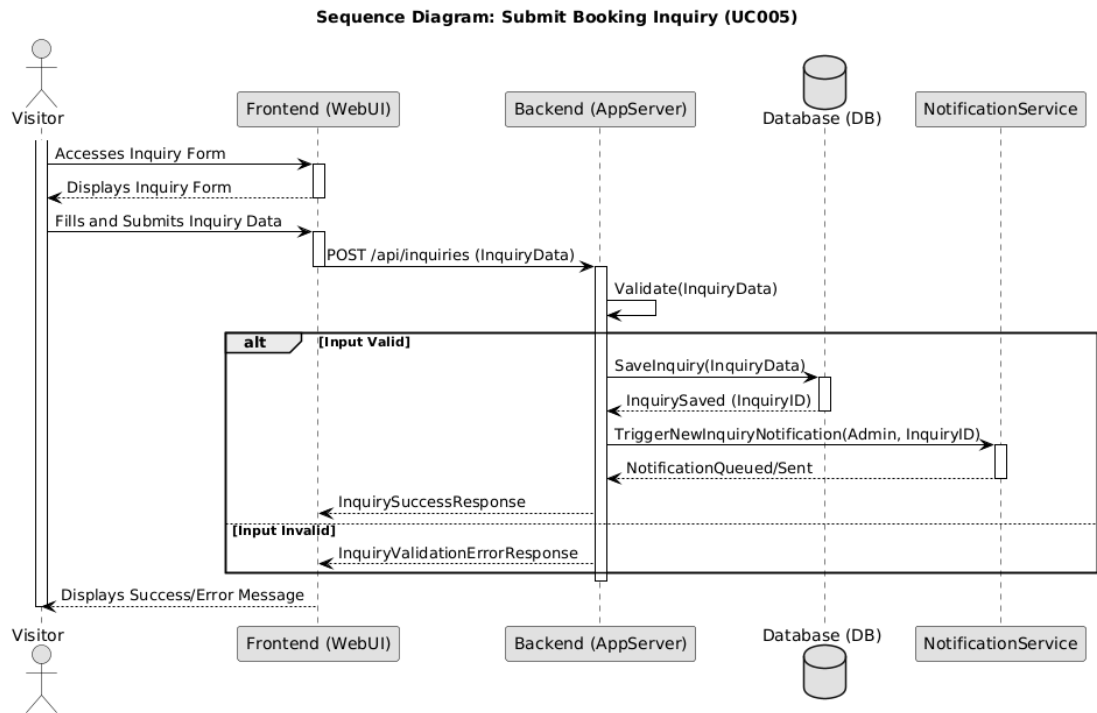


4.3.3.1 Selected Use Case: Submit Booking Inquiry (UC005)

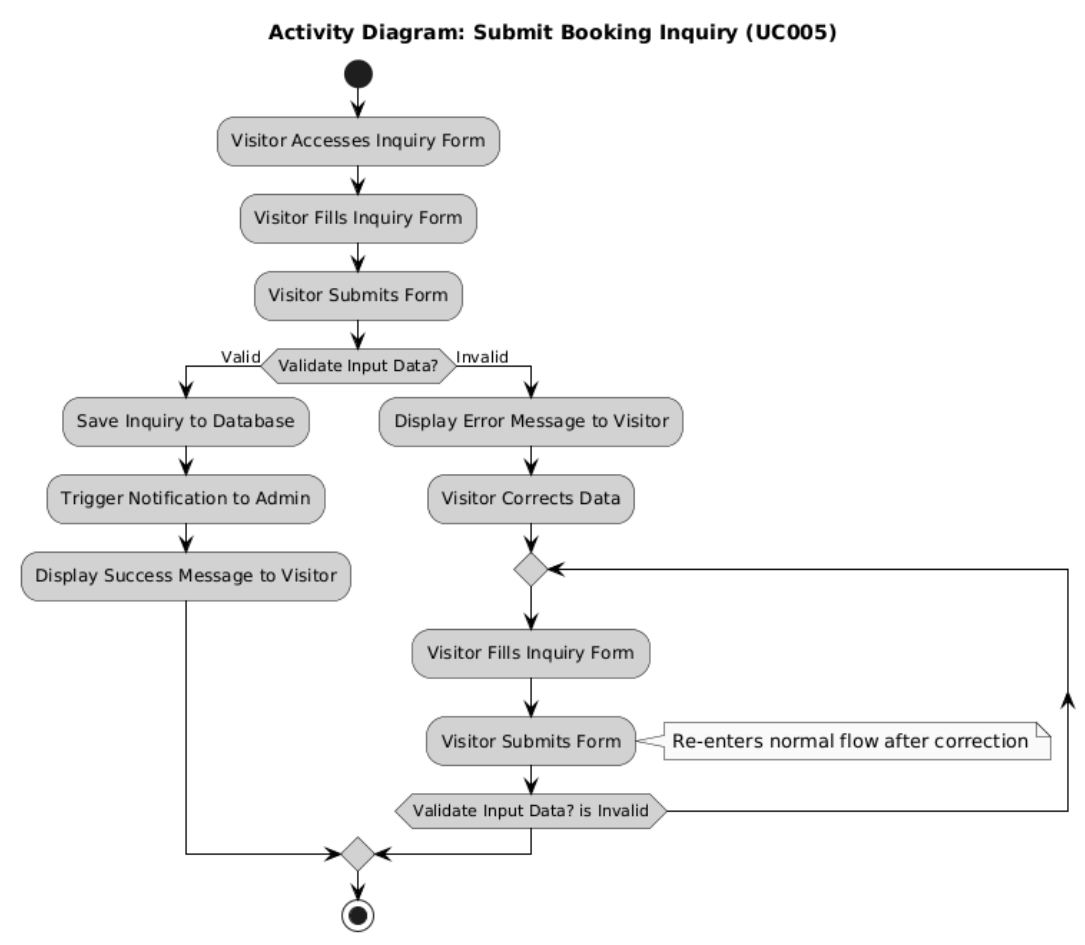
Attribute	Description
Use Case ID	UC005
Use Case Name	Submit Booking Inquiry

Actor(s)	Visitor
Pre-condition(s)	Visitor is on the Al-Muneer Online Portal and has navigated to the inquiry section/form.
Normal Flow	1. Visitor accesses the booking inquiry form. 2. Visitor fills in required fields (e.g., name, contact, desired date, event type, number of guests, specific requests). 3. Visitor submits the form. 4. System validates the input data. 5. System saves the inquiry details to the database. 6. System displays a success message to the Visitor. 7. System triggers a notification to the Administrator about the new inquiry.
Alternative Flow	4a. Invalid Input Data: 4a1. System displays an error message indicating the fields with invalid data. 4a2. Visitor corrects the data and re-submits (back to step 3). 7a. Notification Failure: 7a1. System logs the notification failure but still saves the inquiry and shows success to the visitor. Admin will see the inquiry upon login.
Post-condition(s)	• Booking inquiry is successfully recorded in the system. • Administrator is notified of the new inquiry. • Visitor receives confirmation that their inquiry has been submitted.

4.3.3.2 Sequence Diagram (Submit Booking Inquiry - UC005)



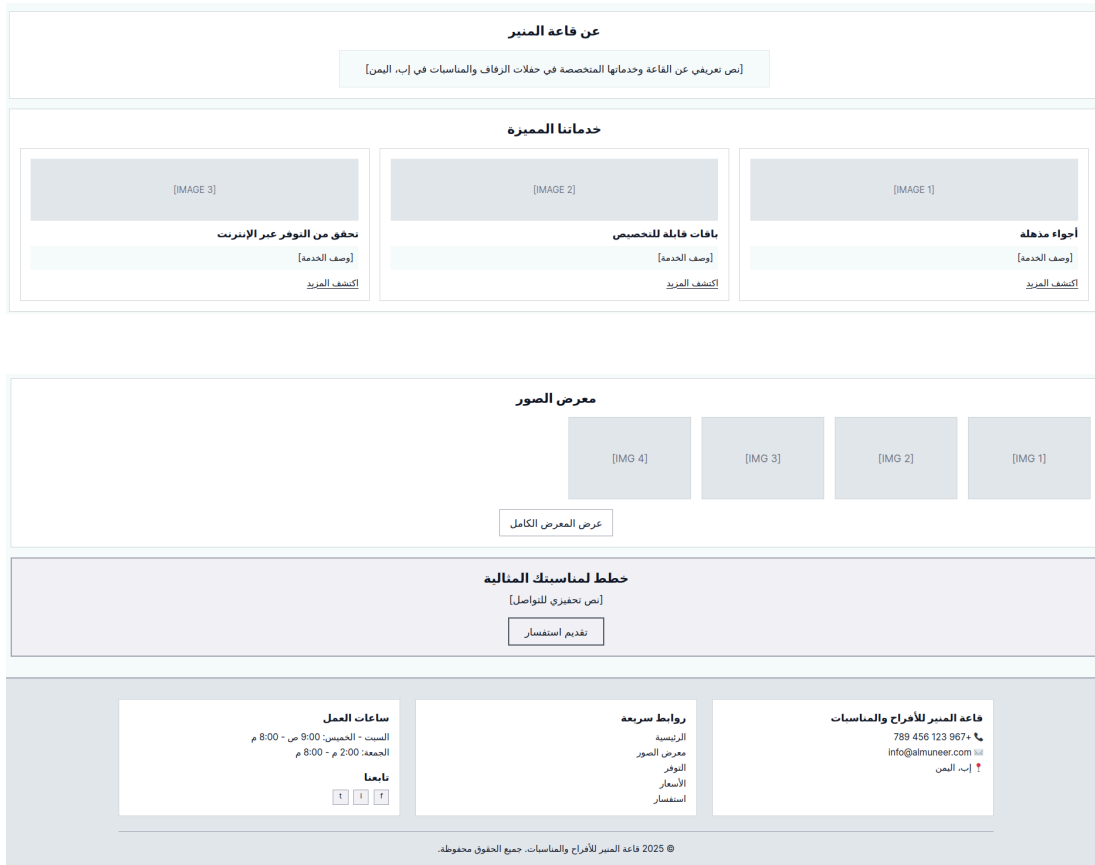
4.3.3.3 Activity Diagram (Submit Booking Inquiry - UC005)



4.4 Interface Design (Low-Fidelity)

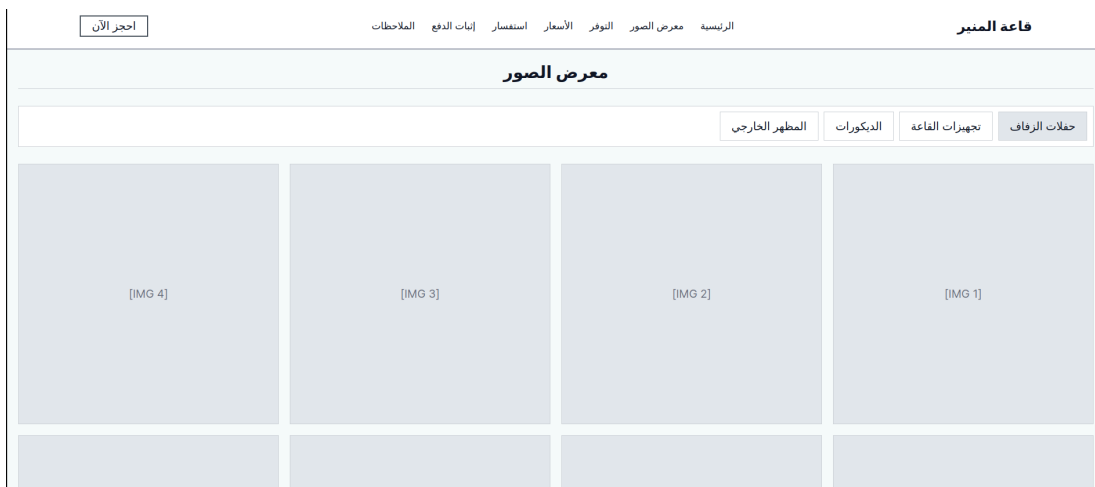
1. Homepage:





- Purpose: Welcome visitors, provide a brief overview of the hall, high-quality hero image/video, clear navigation to other sections (Gallery, Availability, Pricing, Inquiry, FAQ, Feedback).
- Key Elements: Navigation bar, hero section, brief intro text, call-to-action buttons.

2. Gallery Page:



- Purpose: Showcase the venue through images and videos.
- Key Elements: Grid or carousel of media items, ability to view larger images/play videos.

3. Availability Page:

تحقق من التوفر

<
مايو 2025
>

س	ج	خ	ر	ث	ن	ح
7	6	5	4	3	2	1
14	13	12	11	10	9	8
21	20	19	18	17	16	15
28	27	26	25	24	23	22
				31	30	29

محجوز
 قيد الانتظار
 متاح

تقديم استفسار

- Purpose: Allow visitors to check for available dates.
- Key Elements: Interactive calendar highlighting booked/available dates, potentially a date picker to jump to specific months/years.

4. Pricing Page:

الباقات والأسعار

[نص تعريفى عن تنوع الأسعار وإمكانية الحصول على عروض مخصصة]

الباقة الفاخرة

تبدأ من ١,٢٠٠,٠٠٠ ريال يمني

- استخدام القاعة يوم كامل
- تجهيز فاخر
- نظام صوت احترافي
- إضاءة متطورة
- 300 صيف
- خدمة تصوير متكاملة

استفسر عن هذه الباقة

الباقة المميزة

تبدأ من ٨٠٠,٠٠٠ ريال يمني

- استخدام القاعة 6 ساعات
- تجهيز متكامل
- نظام صوت متطور
- إضاءة احترافية
- 200 صيف
- ديكورات زهور

استفسر عن هذه الباقة

الباقة الأساسية

تبدأ من ٥٠٠,٠٠٠ ريال يمني

- استخدام القاعة 4 ساعات
- تجهيز أساسي
- نظام صوت أساسي
- إضاءة قياسية
- 100 صيف

استفسر عن هذه الباقة

خدمات إضافية

خدمة تصوير إضافية
[السعر]

عروض زفاف خاصة
[السعر]

خدمة تقديم الطعام
[السعر]

خدمة صياقة VIP
[السعر]

خدمة موسيقى حية
[السعر]

ديكورات إضافية
[السعر]

احصل على عرض سعر مخصص

- Purpose: Provide clear information about pricing structures and packages.
- Key Elements: Sections for different packages/services, details of what's included, base prices.
- Note: These are dummy pricings and not accurate. The stakeholder will have an access to handle and set all the pricings.

5. Inquiry Page:

تقديم طلب حجز

- 6. Payment Proof Upload Page (May be part of inquiry follow-up or a separate link):**

إرفاق إثبات الدفع

الرجاء إرفاق صورة واضحة أو لقطة شاشة لإيصال تحويل الدفع الخاص بك.
الصيغ المقبولة: JPG, PNG. الحد الأقصى: 5 ميجابايت.

رقم طلب الحجز

أدخل رقم طلب الحجز

اسمك

أدخل اسمك الكامل

إثبات الدفع



اضغط لاختيار ملف
أو اسحب وأفلت الملف هنا

ملاحظات إضافية (اختياري)

أدخل أي ملاحظات إضافية

🔒 جميع المعلومات المرسلة آمنة ومشفرة

رفع الإثبات

- Purpose: Allow users to upload their payment proof securely.
- Key Elements: File upload input, instructions on accepted formats/sizes, submit button.

7. Feedback Page:

شاركنا رأيك

نرحب بملاحظاتك وآرائك لمساعدتنا في تحسين خدماتنا.

اسمك (اختياري)
أدخل اسمك

رقم الهاتف/البريد الإلكتروني (اختياري)
أدخل وسيلة اتصال

موضوع الملاحظات (اختياري)
أدخل موضوع الملاحظات

ملاحظاتك / رسالتك
أدخل ملاحظاتك أو رسالتك

التقييم العام (اختياري)

★

★

★

★

★

إرسال الملاحظات

- Purpose: Collect feedback from visitors.
- Key Elements: Form for name (optional), contact (optional), feedback text, rating (optional), submit button.

4.5 Chapter Summary

This chapter defined the functional and non-functional requirements that will govern the development of the Al-Muneer Online Portal. It detailed the system's layered architecture, encompassing the frontend, backend, and data tiers, and presented a conceptual database design (ERD) to manage the portal's data. Key user interactions were illustrated through a use case diagram, with a specific use case ("Submit Booking Inquiry") further detailed via a flow description, sequence

diagram, and activity diagram to clarify its process. Finally, conceptual ideas for the user interface design of major pages were outlined, aiming for usability and alignment with the project's objectives. These requirements and design considerations will serve as the foundation for the subsequent development phases in PSM2 and will be formally documented in the SRS and SDD.

CHAPTER 5

CONCLUSION

5.1 Introduction

This chapter reflects on the achievements made during the PSM1 phase for the Al-Muneer Online Portal project and outlines the immediate areas for subsequent development in PSM2. The project was designed to address the inefficiencies of manual inquiry and booking management for Al-Muneer Hall by providing a centralized, user-friendly web-based platform. By defining a clear methodology, conducting thorough requirements analysis, and establishing a foundational system design incorporating features such as an interactive calendar, dynamic pricing display, online inquiry submission, payment proof upload, feedback mechanism, and administrative controls, the system aims to provide an efficient and modern solution for both the venue owner and potential clients. This chapter evaluates the progress made in PSM1 towards meeting the initial project objectives and discusses the planned development and testing enhancements for the upcoming PSM2 phase.

5.2 Achievement of Project Objectives

The Al-Muneer Online Portal project was initiated with specific objectives for PSM1 focused on laying a comprehensive groundwork for successful development. The project successfully analyzed the existing manual processes at Al-Muneer Hall and compared them with contemporary online venue management approaches, identifying key areas where a dedicated web portal could provide significant improvements. A structured system development methodology, the Waterfall model, was chosen and detailed, providing a clear roadmap for the project

lifecycle. The requirements elicitation process led to a well-defined set of functional and non-functional requirements covering visitor interactions and administrative capabilities, including the recently incorporated features for payment proof, feedback, reporting, and notifications. The system architecture was designed with a layered approach, and a conceptual database schema was developed to support the portal's data management needs. Key user interactions were modeled through use cases, sequence diagrams, and activity diagrams, providing clarity on system behavior. Conceptual interface designs were also outlined. This thorough analysis and design work undertaken in PSM1 ensures that the project objectives related to planning and specification have been substantially met, providing a solid foundation for the subsequent implementation phase.

5.3 Future Improvements

As the Al-Muneer Online Portal project transitions from the planning and design stages of PSM1 into the development-focused PSM2 phase, the immediate future work will involve translating the established blueprints into a tangible, functional system. The primary area for development will be the full-stack implementation of the portal based on the Spring Boot framework for the backend and HTML, CSS, and JavaScript for the frontend. This includes building out all specified modules: user authentication for the admin panel, venue information management, media gallery uploads, the interactive availability calendar, dynamic pricing configuration, the booking inquiry system, the simple payment proof upload and verification workflow, the feedback submission and management system, the basic reporting dashboard for the administrator, and the notification system (prioritizing SMS/WhatsApp compatible approaches). Rigorous unit testing will be performed concurrently with development for each module. Following the individual module development, system integration will occur, followed by comprehensive system testing encompassing functional, usability, basic performance, and security aspects, guided by a Software Test Document (STD). The final stage of PSM2 will involve deploying the tested application to the designated Cloud VPS environment

and preparing all final project documentation and reports. Further evaluation of system performance and gathering initial user feedback post-deployment will be crucial to validate its effectiveness in a real-world context.

References

1. Armbrust, M., Fox, A., Griffith, R., Joseph, A. D., Katz, R., Konwinski, A., ... & Zaharia, M. (2010). A view of cloud computing. *Communications of the ACM*, 53(4), 50-58.
2. Buhalis, D., & Law, R. (2008). Progress in information technology and tourism management: 20 years on and 10 years after the Internet—The state of eTourism research. *Tourism management*, 29(4), 609-623.
3. Chaffey, D., & Ellis-Chadwick, F. (2019). *Digital marketing*. Pearson UK.
4. Davenport, T. H., & Harris, J. G. (2007). *Competing on analytics: The new science of winning*. Harvard Business Press.
5. Flanagan, D. (2011). *JavaScript: The definitive guide*. O'Reilly Media, Inc.
6. Fowler, M. (2002). *Patterns of enterprise application architecture*. Addison-Wesley Professional.
7. Garcia-Molina, H., Ullman, J. D., & Widom, J. (2008). *Database systems: The complete book*. Pearson Education India.
8. Gazzaroli, D., Lee, G., & Morrison, A. M. (2019). The impact of third-party online platforms on the European event industry: A focus on Eventbrite. *International Journal of Event and Festival Management*, 10(3), 238-253.
9. Harrigan, P., Ramsey, E., & Ibbotson, P. (2012). Exploring and explaining SME marketing: an investigation of social media adoption. *Journal of Research in Marketing and Entrepreneurship*, 14(2), 177-203.
10. Hayes, D. K., & Miller, A. A. (2011). *Revenue management for the hospitality industry*. John Wiley & Sons.
11. ITU. (2019). *Measuring digital development: Facts and figures 2019*. International Telecommunication Union.
12. Kaplan, A. M., & Haenlein, M. (2010). Users of the world, unite! The challenges and opportunities of Social Media. *Business horizons*, 53(1), 59-68.

13. Kshetri, N. (2013). Privacy and security issues in cloud computing: The role of institutions and institutional evolution. *Telecommunications Policy*, 37(4-5), 372-386.
14. Laudon, K. C., & Laudon, J. P. (2016). *Management information systems: Managing the digital firm*. Pearson.
15. Marcus, A., & Gould, E. W. (2000). Crosscurrents: cultural dimensions and global Web user-interface design. *interactions*, 7(4), 32-46.
16. Morgan, N. A., Vorhies, D. W., & Mason, C. H. (2003). Market orientation, marketing capabilities, and firm performance. *Strategic Management Journal*, 24(8), 699-727.
17. Nielsen, J. (2000). *Designing web usability*. New Riders Publishing.
18. O'Brien, J. A., & Marakas, G. M. (2011). *Management Information Systems*. McGraw-Hill Irwin.
19. Odoom, R., Anning-Dorson, T., & Acheampong, G. (2017). Antecedents of social media usage and performance benefits in small and medium-sized enterprises (SMEs). *Journal of Enterprise Information Management*, 30(3), 383-399.
20. Rescorla, E. (2001). *SSL and TLS: designing and building secure systems*. Addison-Wesley Longman Publishing Co., Inc.
21. Royce, W. W. (1970). Managing the development of large software systems. *Proceedings of IEEE WESCON*, 26(August), 1-9.
22. Schneier, B. (1996). *Applied cryptography: protocols, algorithms, and source code in C*. John Wiley & Sons.
23. Sigala, M., Christou, E., & Gretzel, U. (Eds.). (2012). *Social media in travel, tourism and hospitality: Theory, practice and cases*. Ashgate Publishing, Ltd.
24. Sommerville, I. (2011). *Software engineering*. Pearson.
25. Ukpabi, D. C., & Karjaluoto, H. (2017). Consumers' acceptance of information and communications technology in tourism: A review. *Telematics and Informatics*, 34(5), 618-644.
26. Ververidis, C., & Polyzos, G. C. (2002). Mobile marketing using location based services. *Proceedings of the 1st international conference on Mobile business*.
27. W3C. (2018). Web Content Accessibility Guidelines (WCAG) 2.1. Retrieved from <https://www.w3.org/TR/WCAG21/>

28. Walls, C. (2016). *Spring Boot in Action*. Manning Publications.
29. Werthner, H., & Ricci, F. (2004). E-commerce and tourism. *Communications of the ACM*, 47(12), 101-105.

////skip from here

Appendix A

Software Requirements Specification (SRS)

ChatUTM

Version 1.1

Prepared by: Ahmed Ghaleb

Revision Page

a. Overview

This SRS describes the requirements for **ChatUTM**, an AI chatbot designed to provide **UTM students and staff** with **efficient and structured access** to university-related information. The system supports **eight key use cases**, including authentication, question answering, navigation via suggested questions, response feedback, reporting incorrect answers, language preference selection, chat tab navigation, and exporting responses.

b. Target Audience

- Software developers
- System architects
- Project stakeholders (UTM administration, students, and staff)
- Quality assurance teams

c. Version Control History

Version	Primary Author	Description	Date Completed
1.0	Ahmed Ghaleb	Initial draft of the SRS	[6/02/2024]
1.1	Ahmed Ghaleb	Added 6 other use cases	[8/02/2024]

1. Introduction

1.1 Purpose

The purpose of this **Software Requirements Specification (SRS)** is to define the **functional and non-functional requirements** for **ChatUTM**, an AI-driven chatbot designed to provide **seamless access to public UTM information**. The chatbot aims to **reduce search time and cognitive effort**, ensuring students and staff can efficiently retrieve university-related information in an **interactive conversational format**.

1.2 Scope

ChatUTM is an **AI-powered chatbot** that retrieves and presents official UTM information through:

- **Web scraping and RSS automation** to fetch up-to-date UTM content
- **FAISS-based semantic search** for fast and accurate AI responses
- **Multilingual support** for users preferring different languages
- **User authentication via UTM SSO** for secured access
- **AI-driven conversational query resolution** with relevant references

ChatUTM is accessible through both a **web-based chat interface** and an **optional Telegram bot**.

! Note: ChatUTM will not process personalized or private data. It strictly handles **publicly available university information**.

1.3 Definitions, Acronyms, and Abbreviations

- **AI:** Artificial Intelligence
- **FAISS:** Facebook AI Similarity Search (Vector database for efficient similarity searches)
- **NLP:** Natural Language Processing
- **RSS:** Really Simple Syndication (for automated data ingestion)

1.4 References

- **IEEE Std. 830-1998** – IEEE Recommended Practice for Software Requirements Specifications
- **FAISS Documentation** – <https://faiss.ai>

1.5 Overview

This SRS provides a **structured description** of **ChatUTM**, including:

- **User requirements**
 - **System functionalities**
 - **Use case models**
 - **Performance expectations**
 - **Design constraints**
-

2. Specific Requirements

2.1 User Characteristics

ChatUTM is designed for two primary user groups:

2.1.1 UTM Students and Staff

- Basic knowledge of web and chat applications
- Need **quick and accurate** access to university-related information
- May require information in **multiple languages**

2.2 System Features

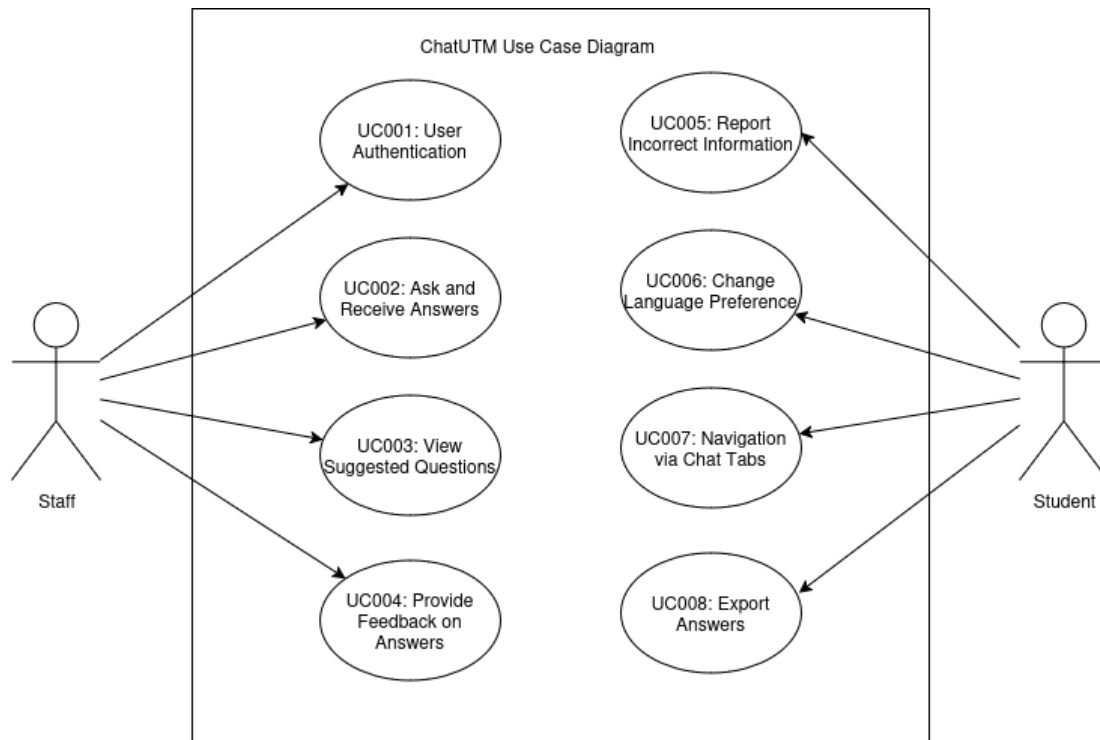
Feature	Description
User Authentication	Users sign in via UTM SSO . Errors are displayed for invalid login attempts.
AI Chat Interface	Users can enter queries and receive AI-generated responses with references.
Suggested Questions	Frequently asked questions are provided for quick access.
Multilingual Support	The chatbot can process and respond in multiple languages .
Data Retrieval	Web scraping and RSS automation keep information up-to-date .
Provide Feedback	Users can rate responses as helpful or unhelpful.
Report Incorrect Information	Users can flag incorrect responses for review.
Chat Tab Navigation	Users can navigate through past conversations using a tab-based system.
Export Responses	Users can copy, generate a PDF, or create a public link for answers.

2.3 Use Case Details

Use Case Table

Use Case ID	Use Case Name	Description	Actors
UC001	User Authentication	Users log in using their UTM credentials, and the system validates authentication.	UTM Students, UTM Staff
UC002	Ask and Receive Answers	Users submit queries through the chat interface, and the system retrieves relevant answers.	UTM Students, UTM Staff
UC003	View Suggested Questions	Users can browse commonly asked questions instead of typing queries.	UTM Students, UTM Staff
UC004	Provide Feedback on Answers	Users can rate responses as helpful or not, aiding system improvement.	UTM Students, UTM Staff
UC005	Report Incorrect Information	Users can flag inaccurate responses for review and correction.	UTM Students, UTM Staff
UC006	Change Language Preference	Users can switch between different languages for better accessibility.	UTM Students, UTM Staff
UC007	Navigation via Chat Tabs	Users can navigate different past conversations via tabs (like ChatGPT).	UTM Students, UTM Staff
UC008	Export Answers	Users can copy responses in a formatted way, generate a PDF, or create a public link for sharing.	UTM Students, UTM Staff

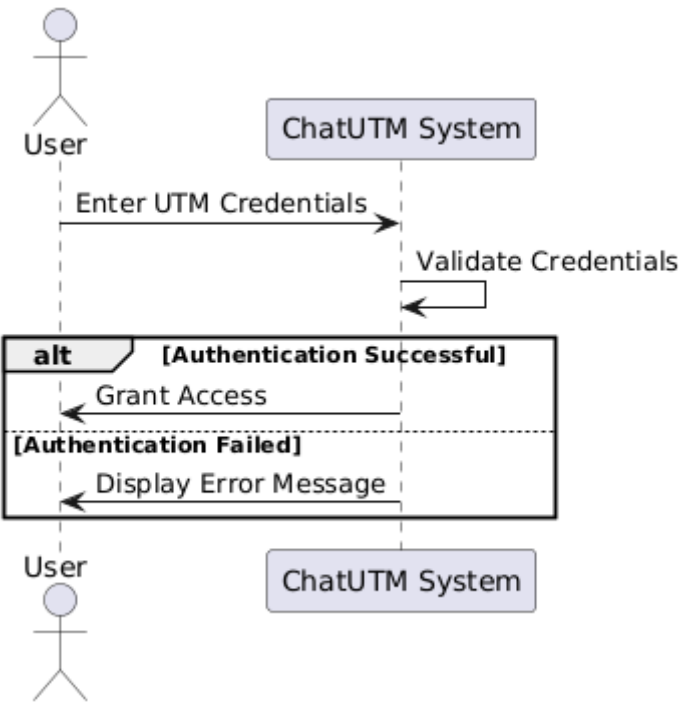
Use Case Diagram



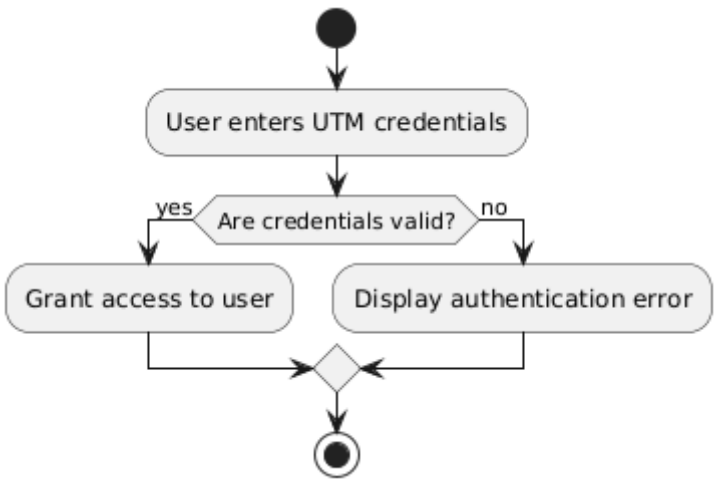
2.3.1 UC001: User Authentication

Attribute	Description
Use Case ID	UC001
Use Case Name	User Authentication
Actor(s)	UTM Students, UTM Staff
Pre-condition(s)	User must have a valid UTM SSO account .
Normal Flow	1. User enters login credentials. 2. System verifies credentials. 3. System grants access or displays an error message.
Alternative Flow	If credentials are invalid, an error message is displayed.
Exception Flow	System failure to verify credentials.
Post-condition(s)	User is authenticated and redirected to the chat interface .

Sequence Diagram (User Authentication)



Activity Diagram (User Authentication)

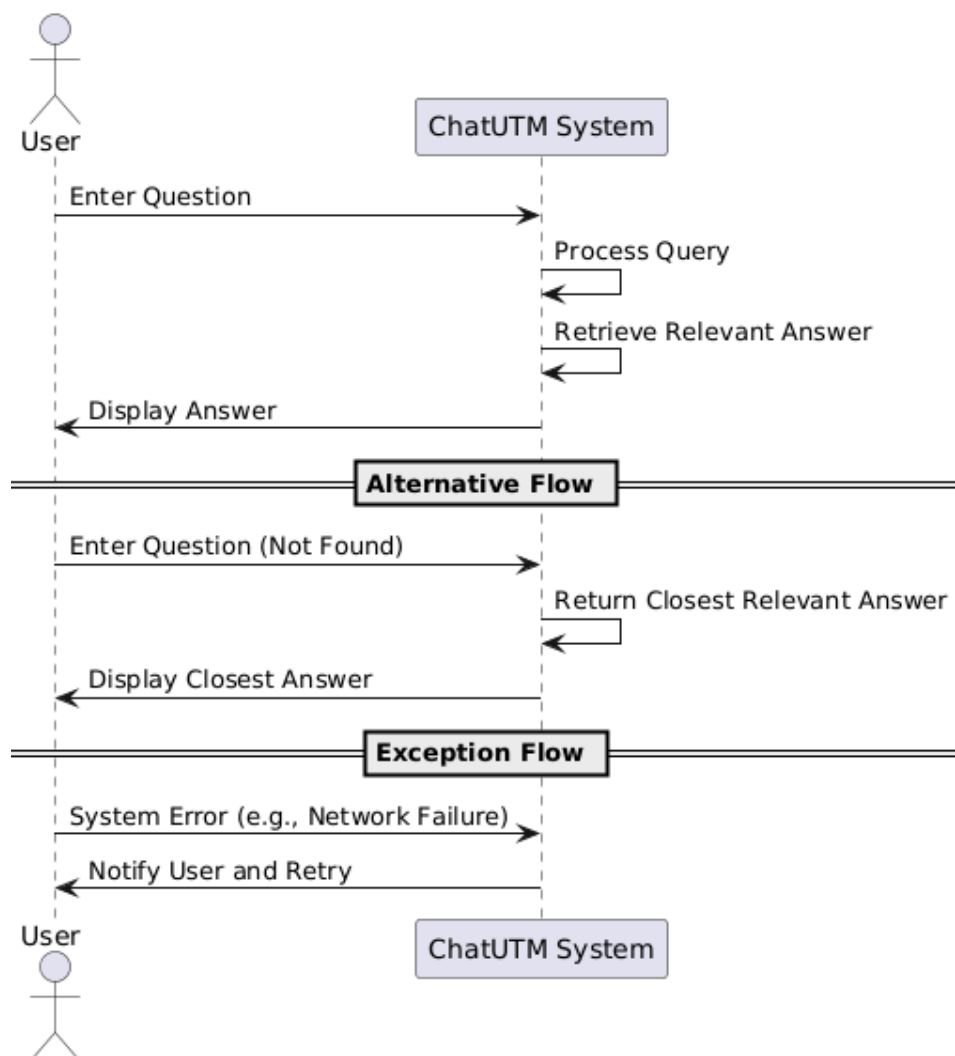


2.3.2 UC002: Ask and Receive Answers

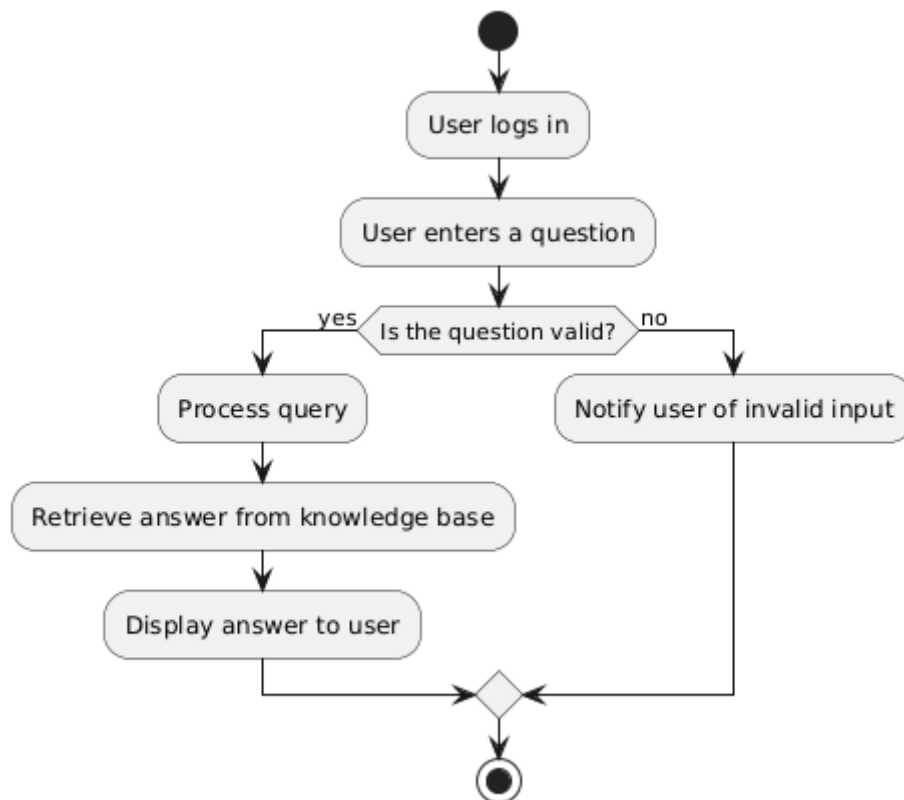
Attribute	Description
Use Case ID	UC002

Use Case Name	Ask and Receive Answers
Actor(s)	UTM Students, UTM Staff
Pre-condition(s)	User is authenticated.
Normal Flow	1. User enters a query in the chat. 2. AI generates a response with relevant references. 3. System displays the response.
Alternative Flow	If no relevant data is found, a fallback message is displayed.
Post-condition(s)	User receives an answer or a guidance message .

Sequence Diagram (Ask and Receive Answers)



Activity Diagram (Ask and Receive Answers)

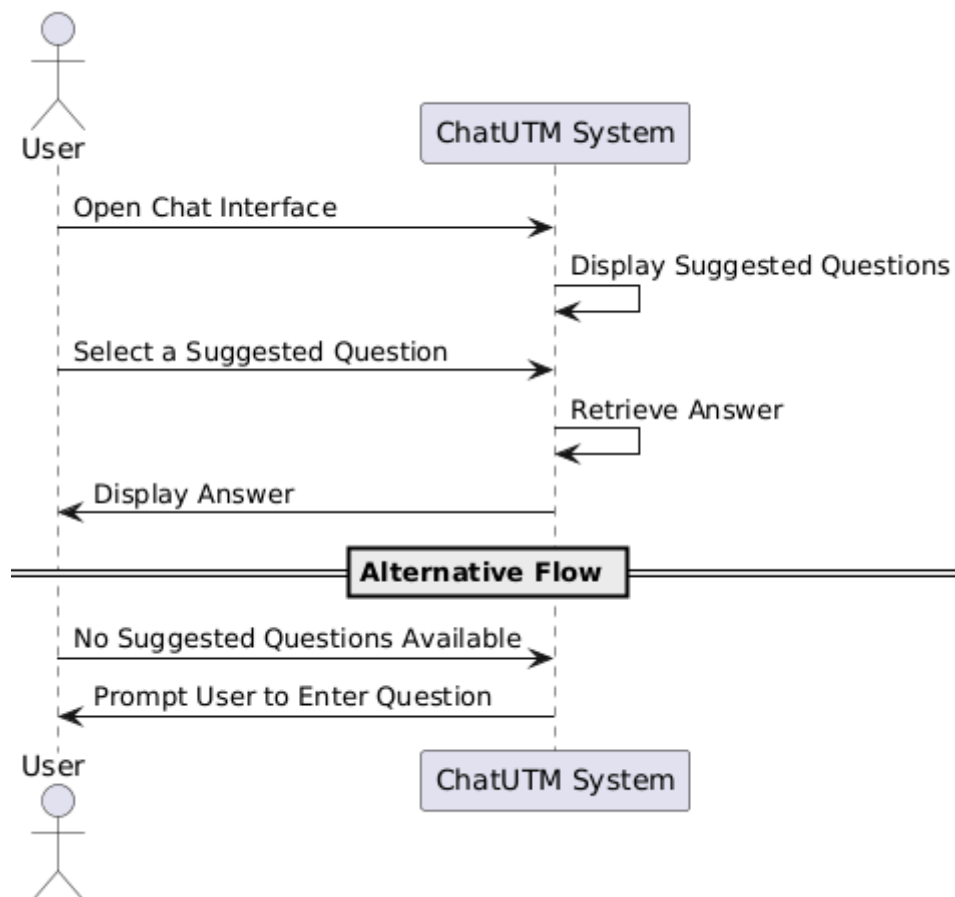


2.3.3 UC003: View Suggested Questions

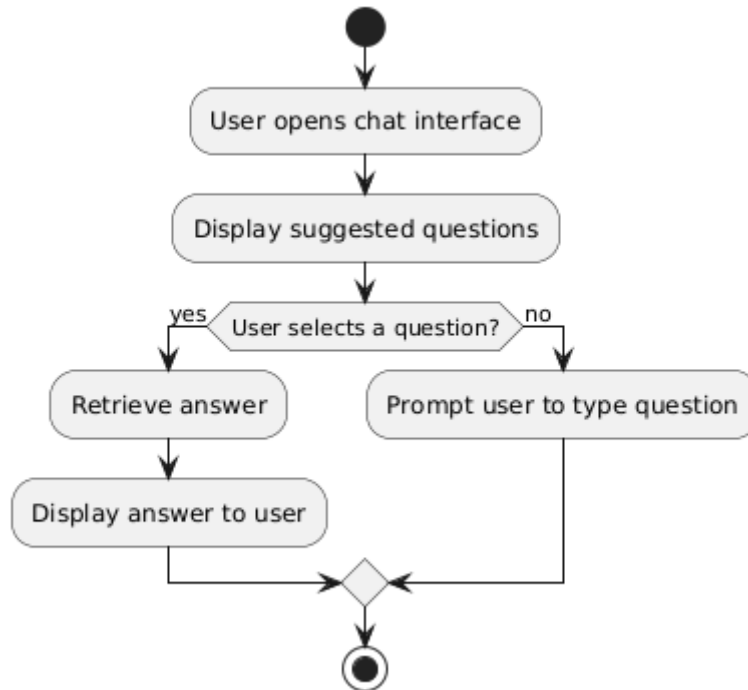
Attribute	Description
Use Case ID	UC003
Use Case Name	View Suggested Questions
Actor(s)	UTM Students, UTM Staff
Pre-condition(s))	User is authenticated.
Normal Flow	1. User opens the chatbot. 2. System displays a list of commonly asked questions . 3. User selects a question from the list.

	4. AI retrieves the most relevant response and displays it.
Alternative Flow	If no predefined questions are available, the user is prompted to type a query manually.
Post-condition(s)	User receives an answer without manually entering a question.

Sequence Diagram (View Suggested Questions)



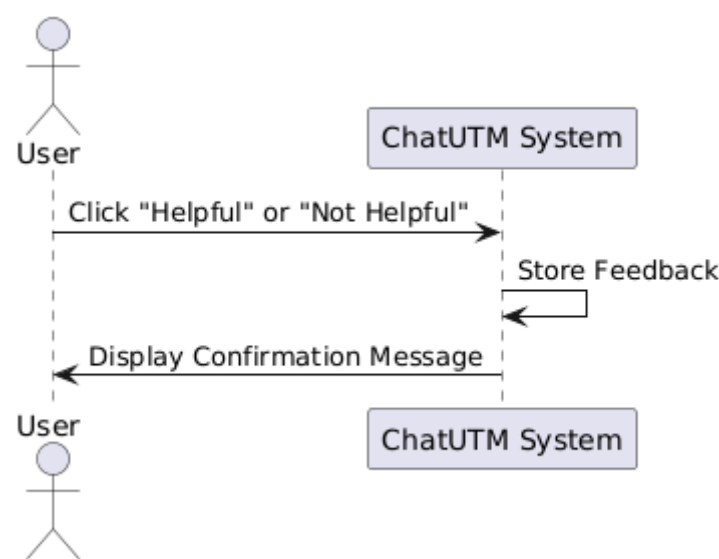
Activity Diagram (View Suggested Questions)



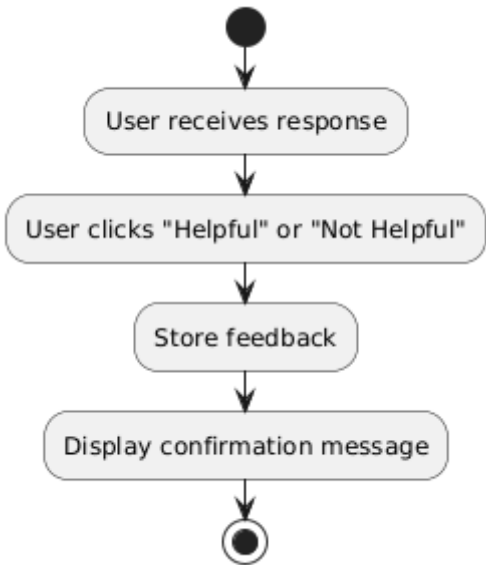
2.3.4 UC004: Provide Feedback on Answers

Attribute	Description
Use Case ID	UC004
Use Case Name	Provide Feedback on Answers
Actor(s)	UTM Students, UTM Staff
Pre-condition(s)	User is authenticated and has received an AI-generated response.
Normal Flow	1. User clicks on "Helpful" or "Not Helpful" for a response. 2. System records the feedback .
Alternative Flow	If the user does not submit feedback, the session continues normally.
Post-condition(s))	The system logs the feedback for future improvements.

Sequence Diagram (Provide Feedback on Answers)



Activity Diagram (Provide Feedback on Answers)



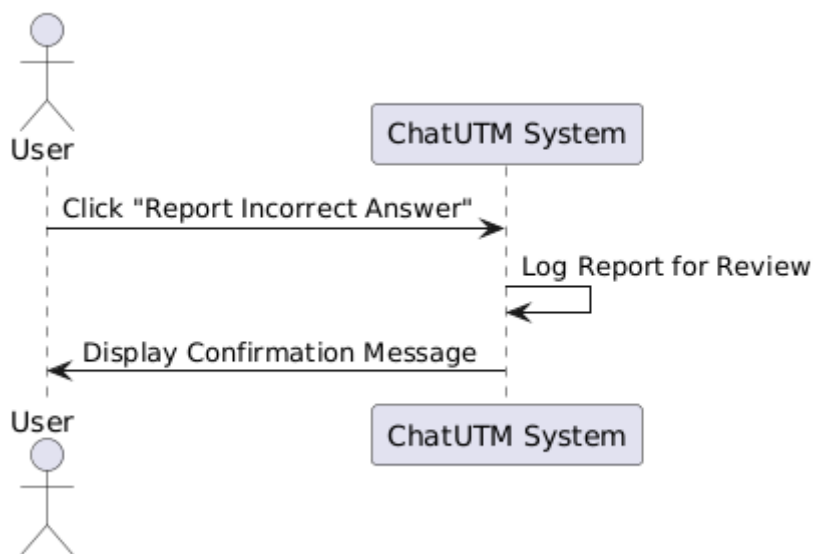
2.3.5 UC005: Report Incorrect Information

Attribute	Description
Use Case ID	UC005
Use Case Name	Report Incorrect Information

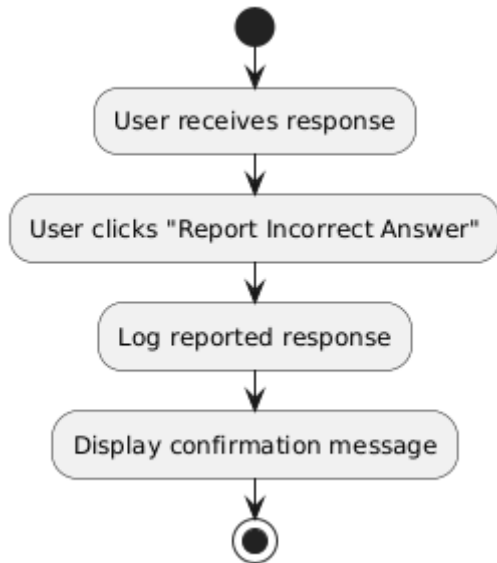
Actor(s)	UTM Students, UTM Staff
Pre-condition(s)	User is authenticated and has received an AI-generated response.
Normal Flow	1. User selects "Report Incorrect Answer". 2. System logs the reported response for review.
Alternative Flow	If the user does not report , the conversation proceeds as usual.

Post-condition(s) System **flags the response for review and potential correction.**

Sequence Diagram (Report Incorrect Information)



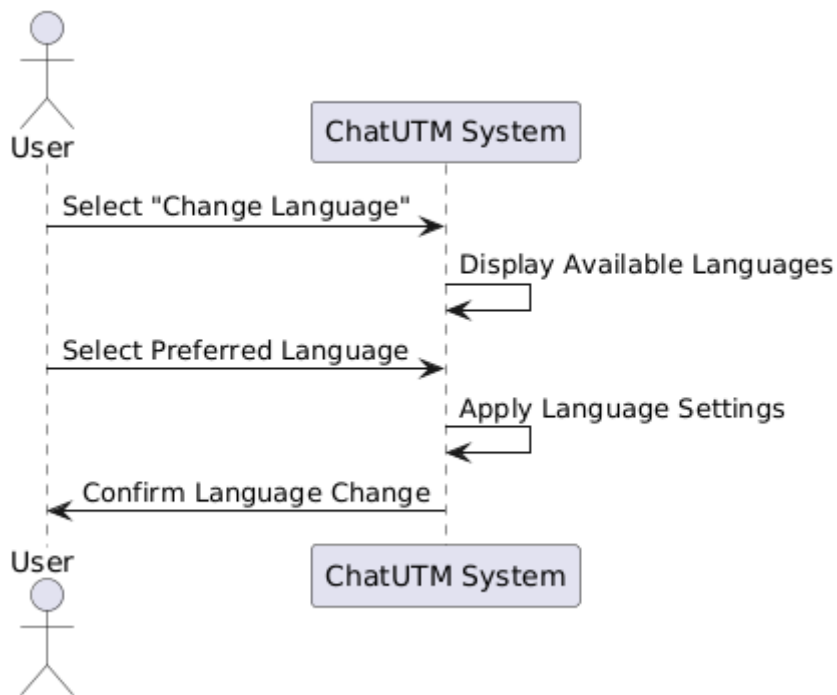
Activity Diagram (Report Incorrect Information)



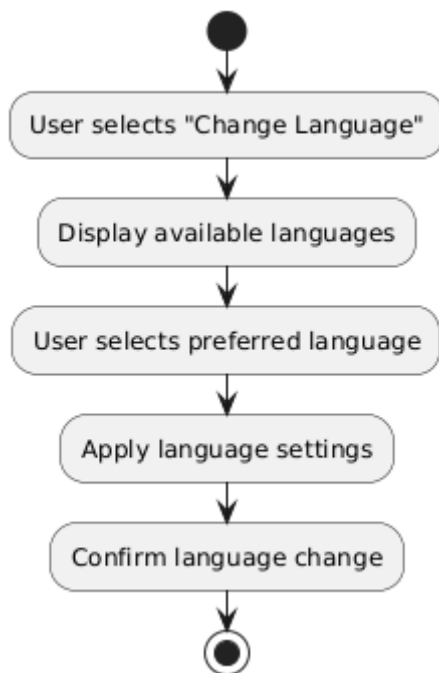
2.3.6 UC006: Change Language Preference

Attribute	Description
Use Case ID	UC006
Use Case Name	Change Language Preference
Actor(s)	UTM Students, UTM Staff
Pre-condition(s)	User is authenticated.
Normal Flow	1. User selects " Change Language " option. 2. System displays available languages . 3. User selects a language . 4. System switches response language .
Alternative Flow	If the selected language is unavailable, the system defaults to English .
Post-condition(s))	System continues responding in the new selected language .

Sequence Diagram (Change Language Preference)



Activity Diagram (Change Language Preference)

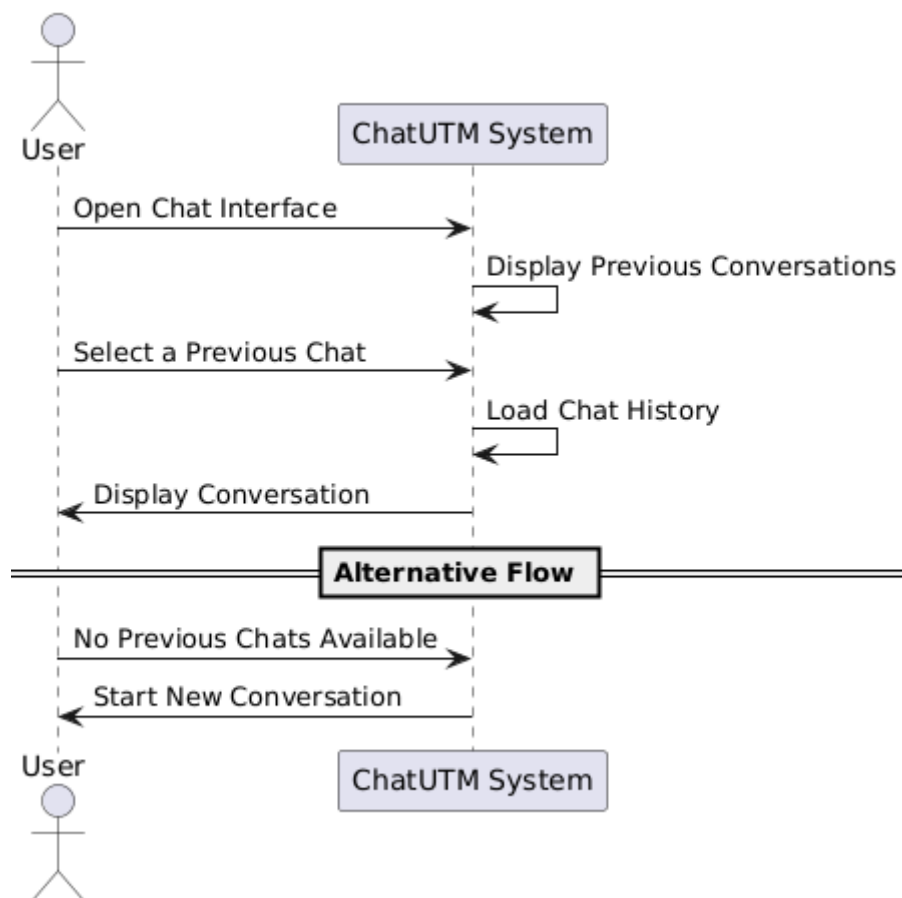


2.3.7 UC007: Navigation via Chat Tabs

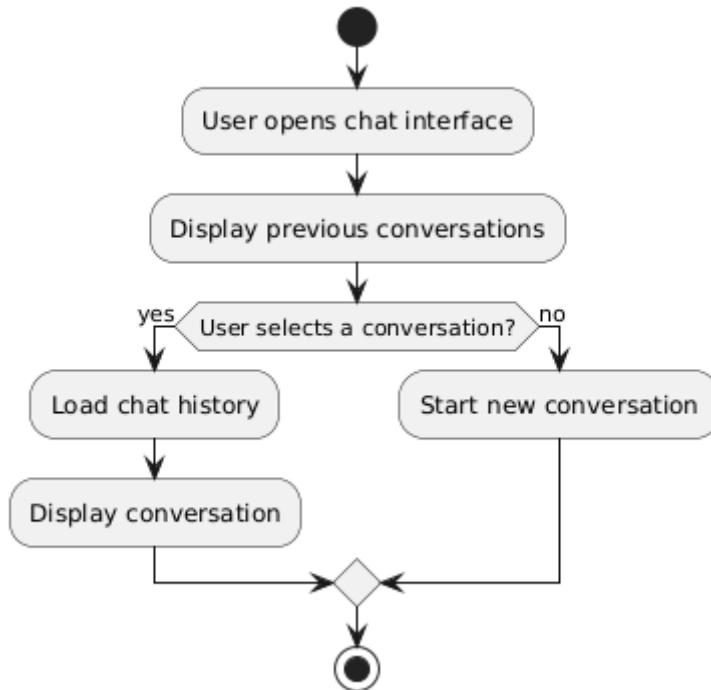
Attribute	Description
-----------	-------------

Use Case ID	UC007
Use Case Name	Navigation via Chat Tabs
Actor(s)	UTM Students, UTM Staff
Pre-condition(s)	User is authenticated.
Normal Flow	1. User selects a previous conversation tab . 2. System loads the past conversation history .
Alternative Flow	If no previous chat exists, the system starts a new conversation .
Post-condition(s)	User can navigate between past and current conversations.

Sequence Diagram (Navigation via Chat Tabs)



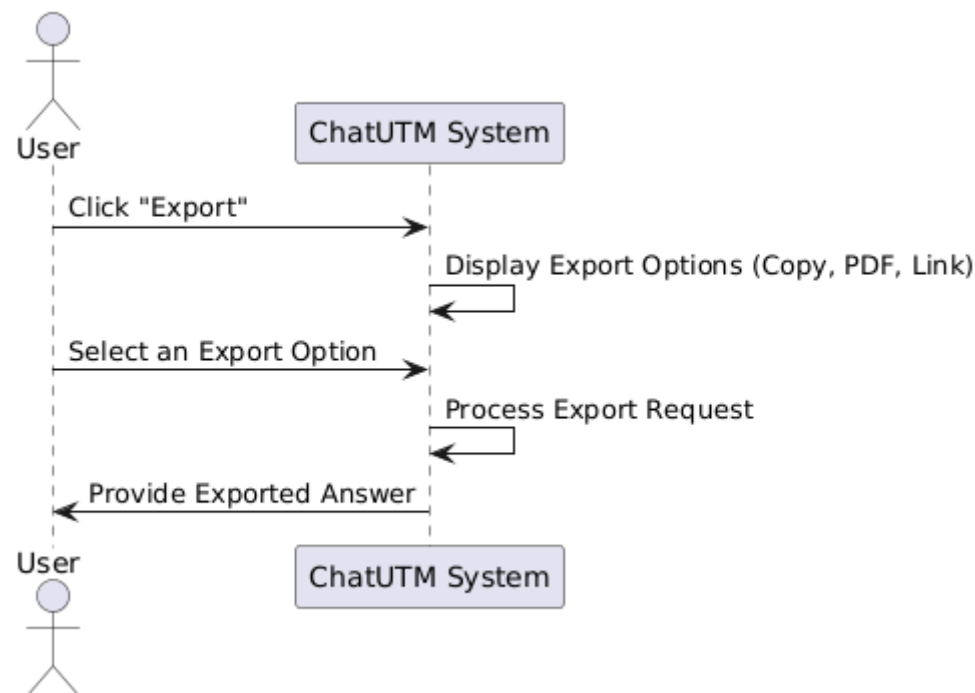
Activity Diagram (Navigation via Chat Tabs)



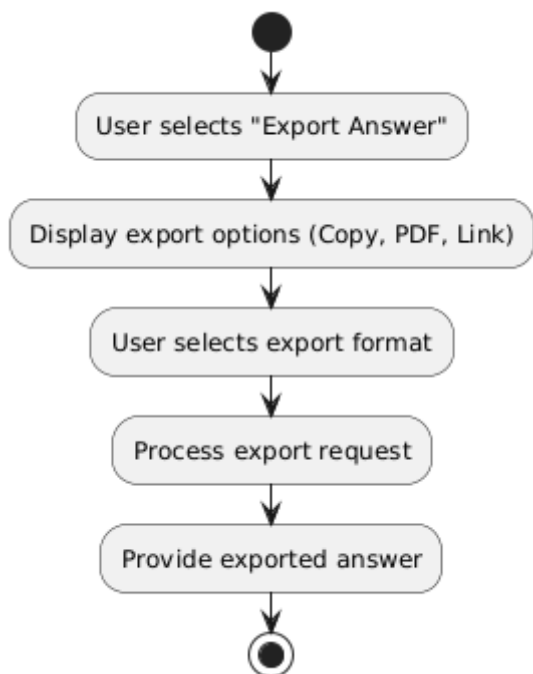
2.3.8 UC008: Export Answers

Attribute	Description
Use Case ID	UC008
Use Case Name	Export Answers
Actor(s)	UTM Students, UTM Staff
Pre-condition(s)	User is authenticated and has received an AI-generated response.
Normal Flow	1. User selects "Copy", "Generate PDF", or "Share Link". 2. System exports the response in the chosen format .
Alternative Flow	If export fails, an error message is displayed.
Post-condition(s)	User can download or share the answer externally.

Sequence Diagram (Export Answers)



Activity Diagram (Export Answers)



2.4 Performance and Other Requirements

- **Response Time:** within **3-10 seconds** for 90% of queries.
- **Availability:** 99.9% uptime.
- **Scalability:** Docker-based deployment for cloud and on-prem scalability.
- **Security:** OAuth-based authentication via UTM SSO.

2.5 Design Constraints

- The system **must adhere** to UTM data privacy policies.
 - Only **public UTM data** is stored and processed.
 - **Multilingual support** must include English and Malay at a minimum.
-

Software Design Specification (SDS) for ChatUTM

Version 1.0

Prepared by: Ahmed Hani Ghaleb

Faculty of Computing, Universiti Teknologi Malaysia

1. Revision Page

1.1 Overview

This document provides the **Software Design Specification (SDS) for ChatUTM**, an **AI-driven chatbot** designed to streamline information retrieval for UTM students and staff. The document outlines the **architectural design, system components, design decisions, and workflows** to ensure a **scalable, efficient, and secure system**.

1.2 Issuing Organization / Team

- **Organization:** Universiti Teknologi Malaysia (UTM)
- **Project Team:** Individual project by **Ahmed Ghaleb**

1.3 Authorship / Project Team Members

Name	Role	Task Status
Ahmed Ghaleb	Developer	In Progress
Ahmed Ghaleb	System Architect	Complete

Ahmed Ghaleb	Documentation	In Progress
--------------	---------------	-------------

1.4 Revision History

Revision No.	Issue Date	Details of Revision
00	02/01/2025	Initial Submission
01	08/02/2025	Updates based on feedback

2. Introduction

2.1 Purpose

This SDS serves as a **technical blueprint** for the **ChatUTM AI chatbot**, detailing its **architecture, workflows, components, and interactions**. The intended audience includes:

- **Developers** (for system implementation and maintenance)
- **System Architects** (for design improvements)
- **Project Stakeholders** (for understanding system capabilities)

2.2 Scope

ChatUTM is a **university-wide knowledge assistant** that:

- ✓ **Consolidates** UTM public data from **websites, portals, RSS feeds, and Telegram**.
- ✓ **Uses AI and semantic search** (FAISS, DeepSeek embeddings) to provide **fast, relevant answers**.
- ✓ **Supports multiple languages** and is **accessible via a web interface and Telegram bot**.
- ✓ **Allows user feedback** on responses, **reporting incorrect answers**, and **exporting responses**.

2.3 Context

ChatUTM is intended for **UTM students and staff** who need **quick access** to university-related information. Instead of manually navigating multiple sources, users interact with ChatUTM’s **AI-driven chat interface** for immediate answers.

2.4 Summary

This SDS provides:

- **System architecture and component design**
 - **Use cases and system workflows**
 - **Security and performance considerations**
-

3. References

- **IEEE Std. 1016-1998** – IEEE Recommended Practice for Software Design Descriptions
 - **FAISS Documentation** – <https://faiss.ai>
 - **DeepSeek Embeddings** – <https://deepseek.ai>
-

4. Glossary

Term	Definition
FAISS	Facebook AI Similarity Search - vector database for efficient similarity-based search.
NLP	Natural Language Processing - AI technique for understanding user queries.
RSS	Really Simple Syndication - automated data ingestion method.
SSO	Single Sign-On - UTM authentication system.

5. Design Body

5.1 Design Stakeholders and Their Concerns

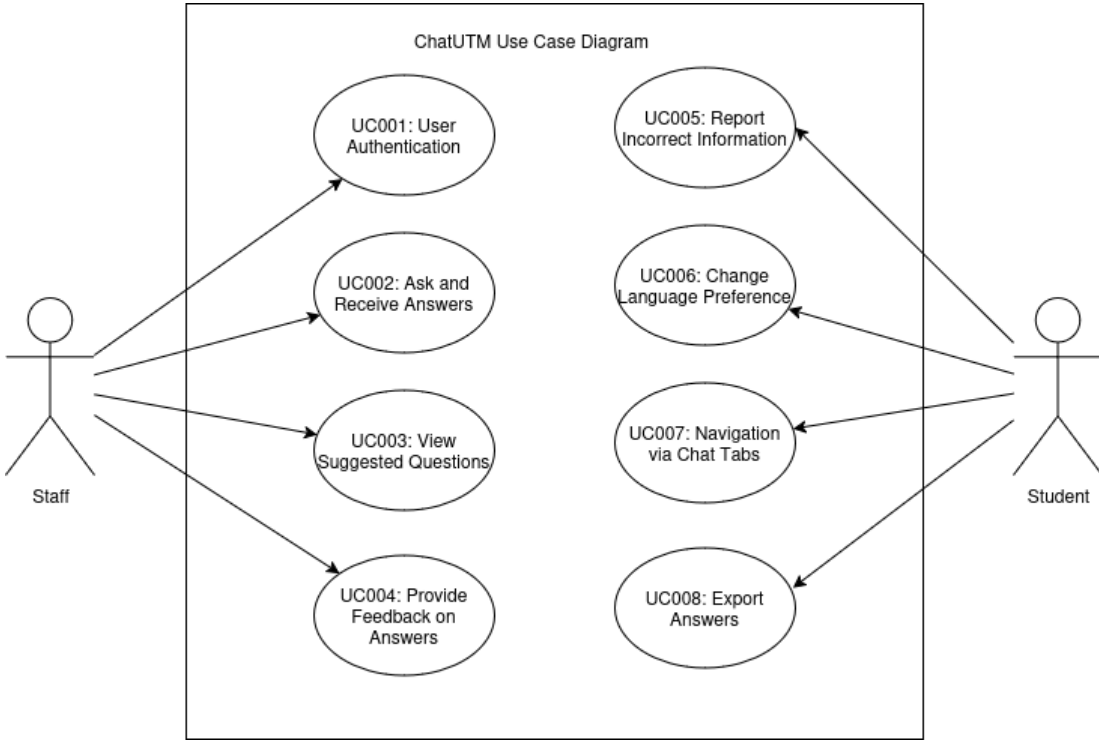
Stakeholder	Concern
Students & Staff	Quick access to university-related information.
UTM IT Team	Security, scalability, and ease of maintenance.
Developers	Clean, modular, and well-documented system architecture.

5.2 Context Viewpoint

5.2.1 Design Concerns

- Ensuring efficient AI-driven response generation.
- User authentication using UTM SSO.
- Multi-platform accessibility (web and Telegram).

5.2.2 Design View (Use Case Diagram Placeholder)

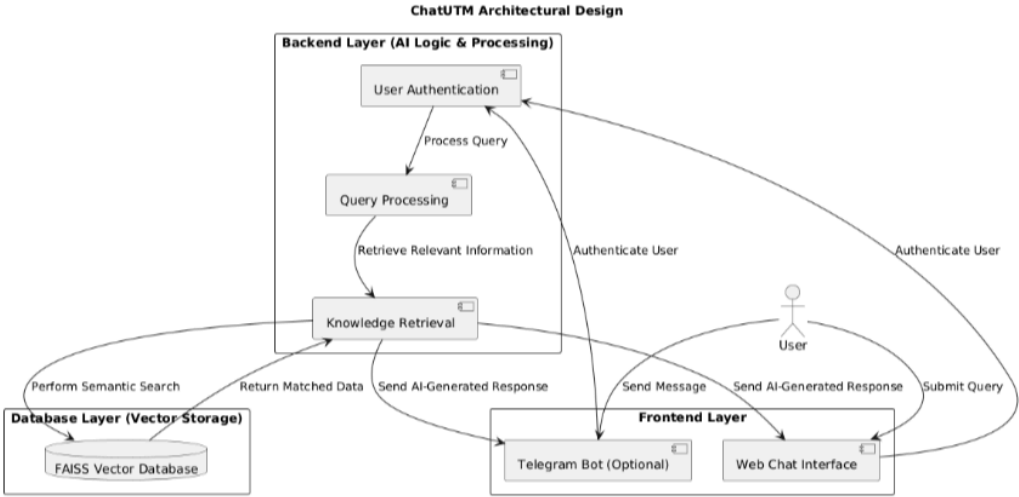


5.3 Composition Viewpoint

5.3.1 Design Concerns

- System is divided into three main layers:
 1. Frontend Layer (Web & Telegram UI)
 2. Backend Layer (Authentication, AI Processing, Knowledge Retrieval)
 3. Database Layer (FAISS Vector Storage for AI embeddings)

5.3.2 Design View (Architecture Diagram Placeholder)

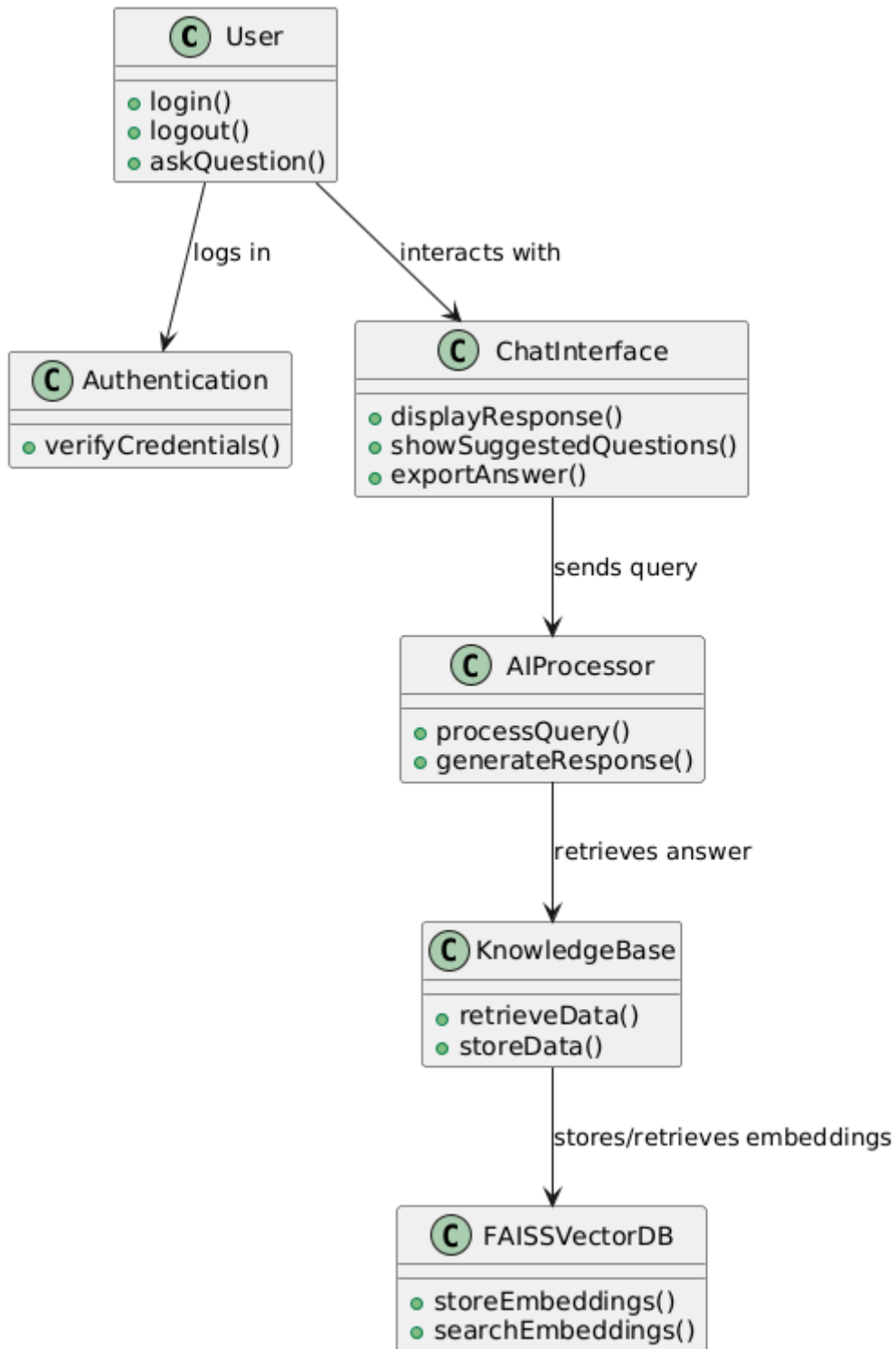


5.4 Logical Viewpoint

5.4.1 Design Concerns

- Class structures for handling authentication, query processing, and AI responses.

5.4.2 Design View (Class Diagram Placeholder)



5.5 Information Viewpoint

5.5.1 Design Concerns

- Data storage in FAISS vector format instead of relational tables.

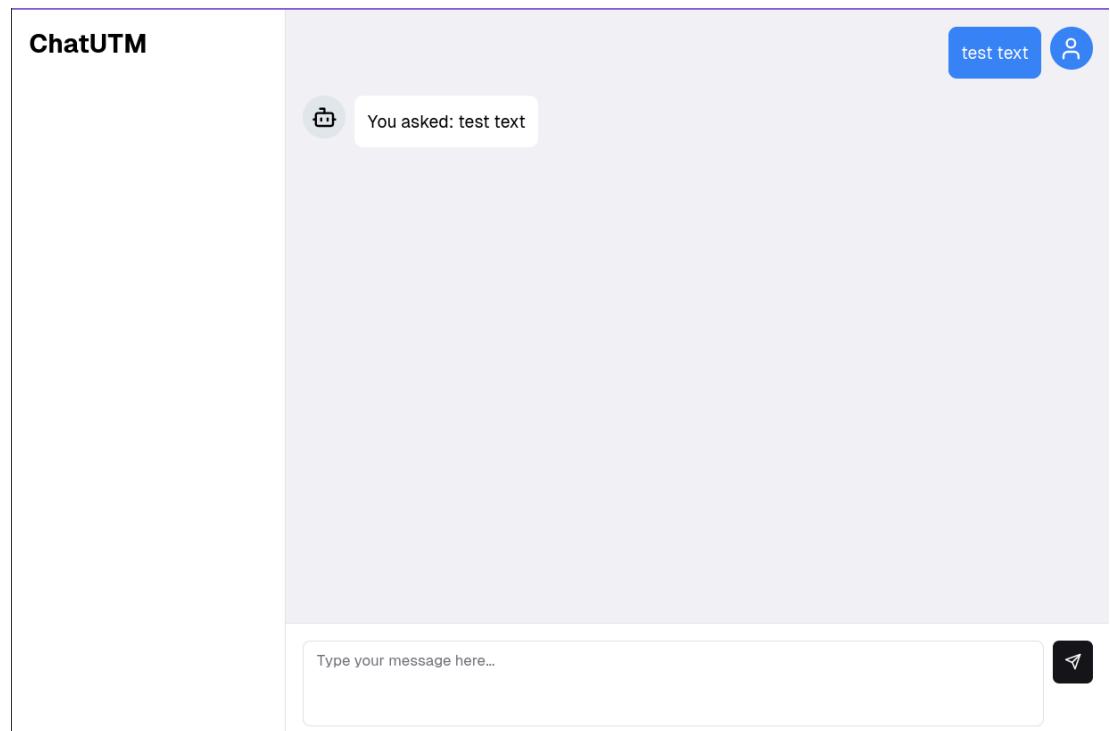
5.6 Interface Viewpoint

5.6.1 Design Concerns

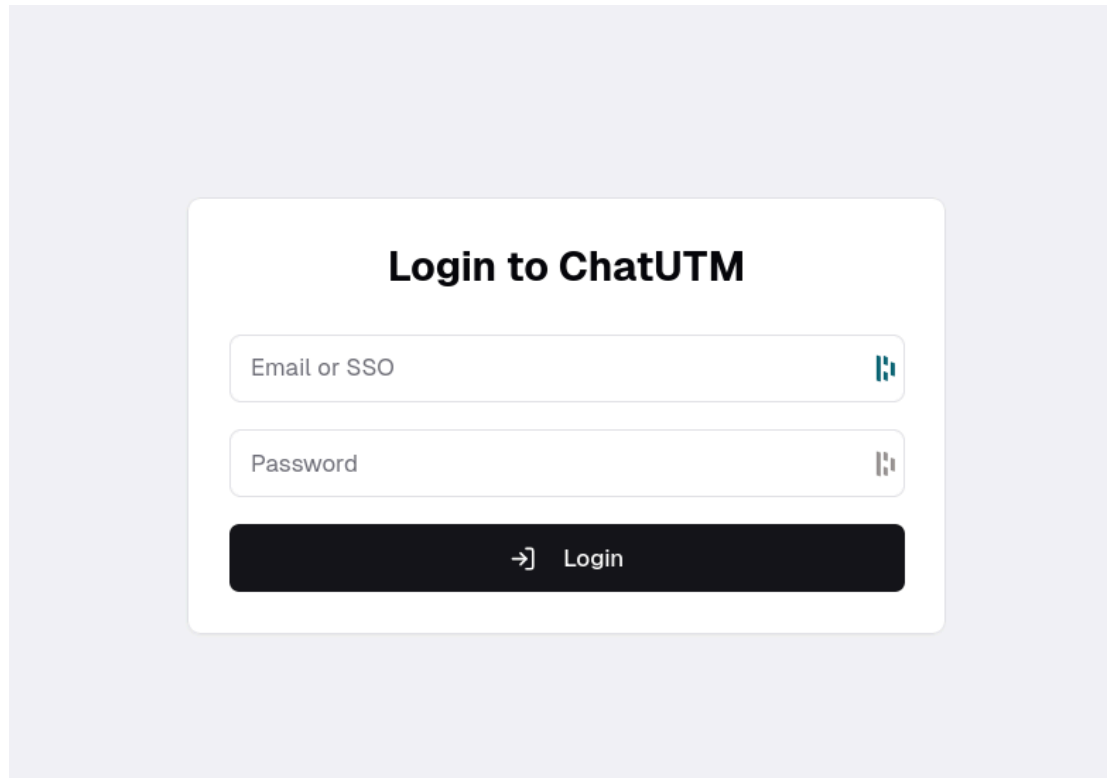
- User-friendly interface with chat-based interactions.
- Multilingual support toggle.

5.6.2 Design View (Interfaces Placeholder)

Chat Page



Login Page

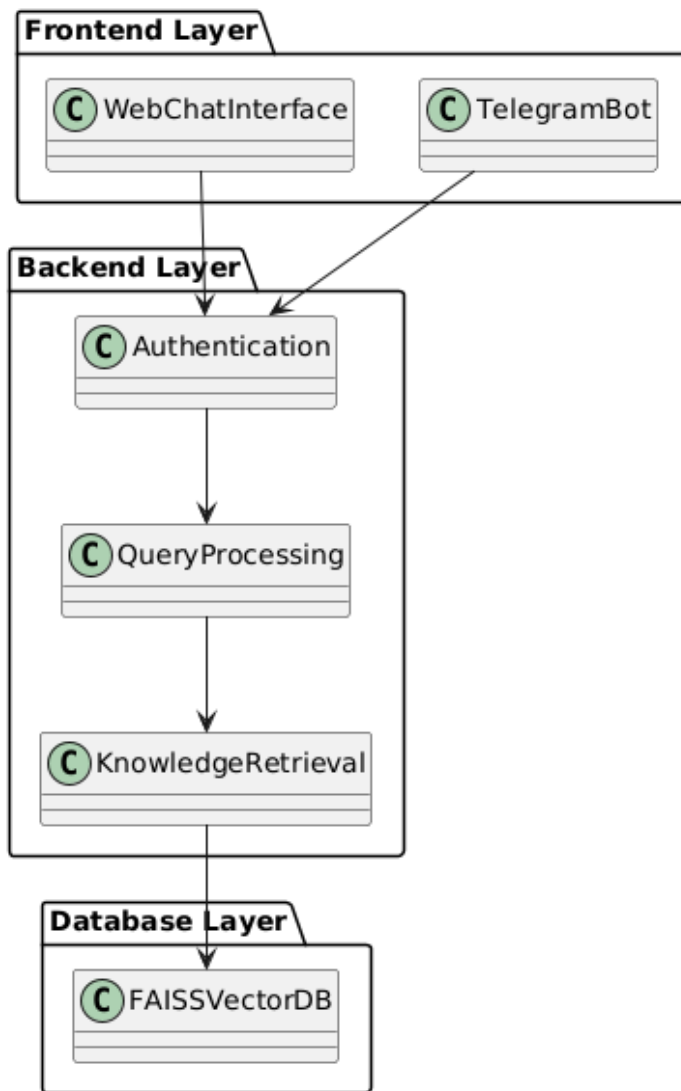


5.7 Structure Viewpoint

5.7.1 Design Concerns

- Separation of concerns between UI, backend processing, and data retrieval.

5.7.2 Design View (Package Diagram Placeholder)



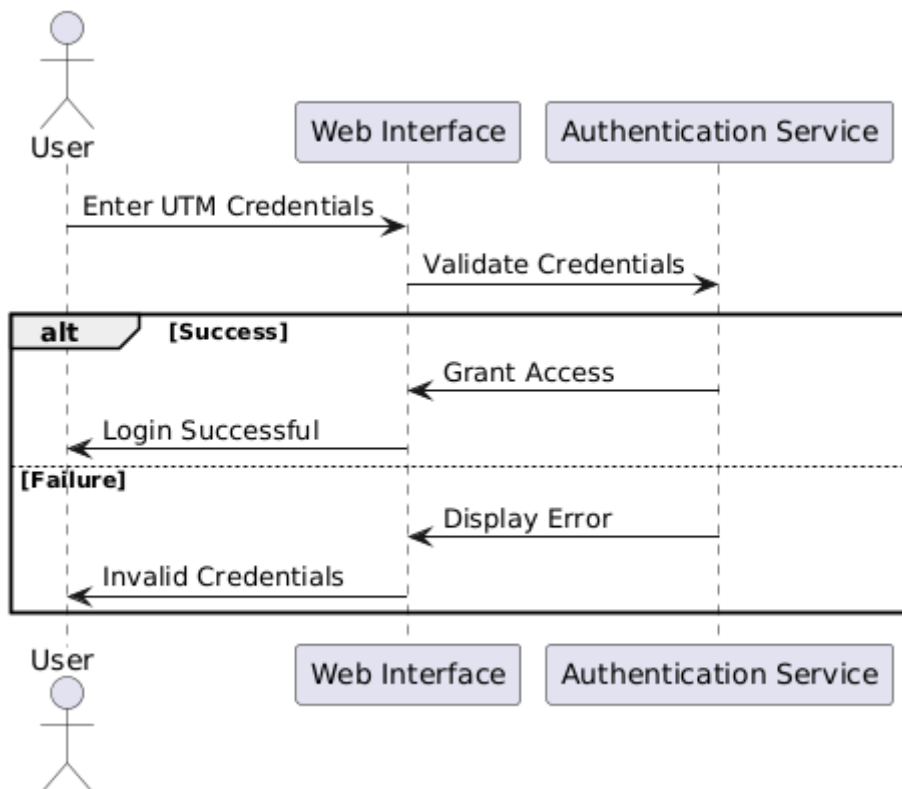
5.8 Interaction Viewpoint

5.8.1 Design Concerns

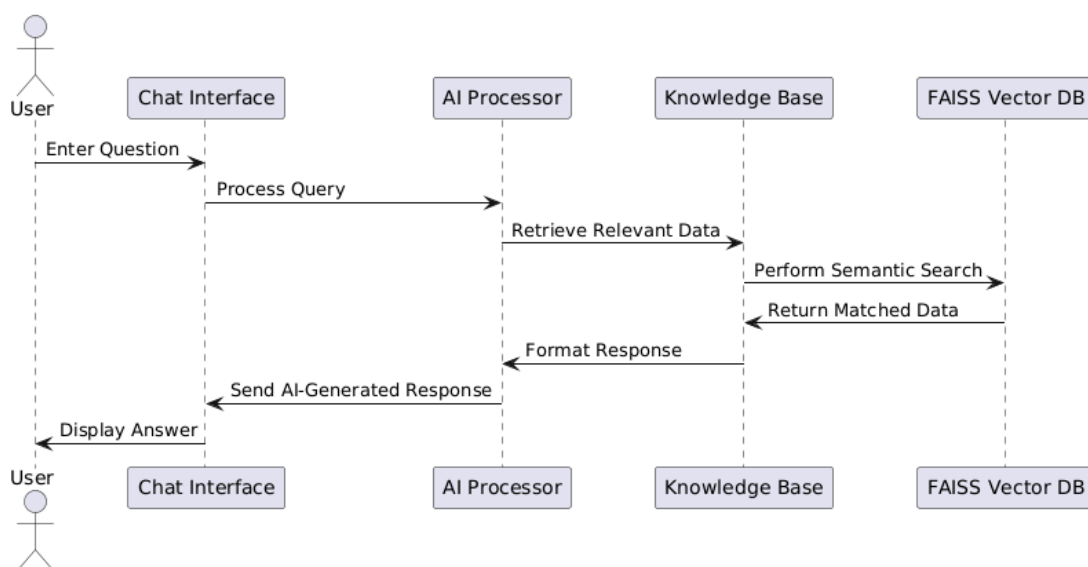
- Sequence of interactions between users and system components.

5.8.2 Design View (Sequence Diagrams Placeholder)

Sequence Diagram (User Authentication)



Sequence Diagram (Ask and Receive Answers)

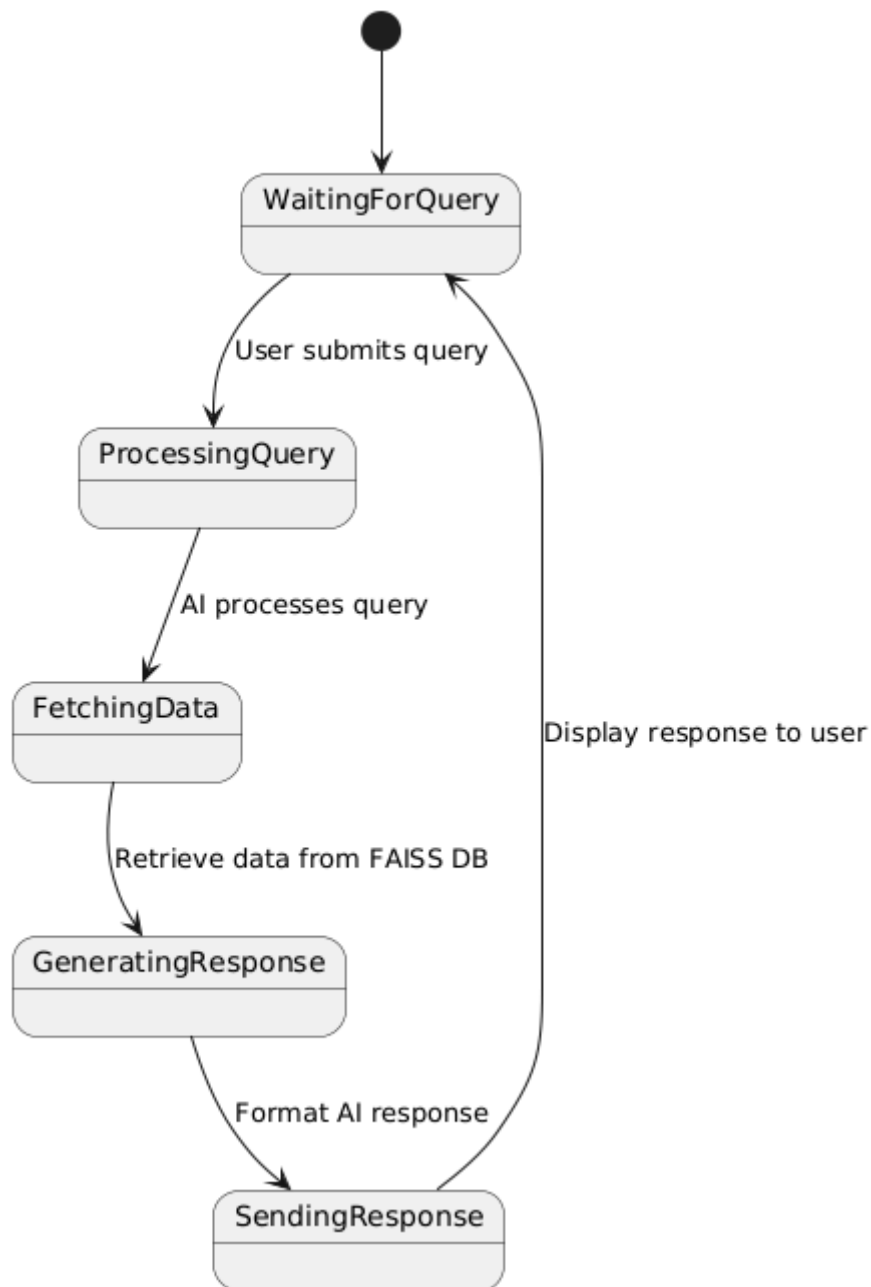


5.9 State Dynamic Viewpoint

5.9.1 Design Concerns

- Handling different chatbot states (waiting for query, processing, responding, feedback submission).

5.9.2 Design View (State Machine Diagram Placeholder)

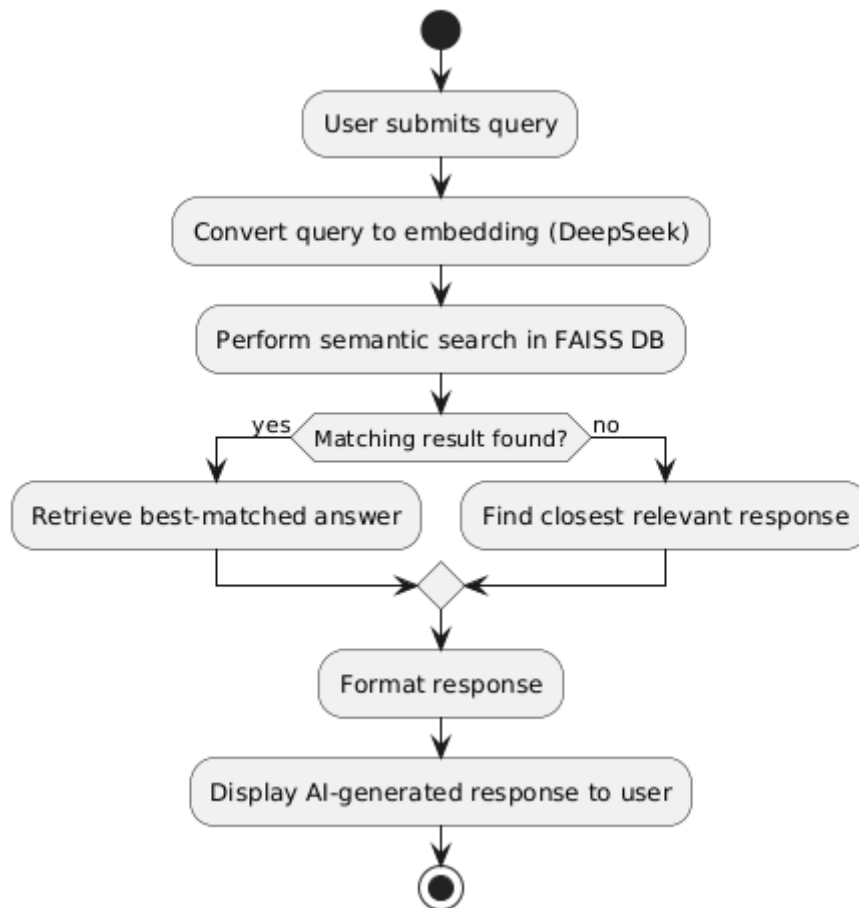


5.10 Algorithm Viewpoint

5.10.1 Design Concerns

- AI-driven query handling and retrieval efficiency.

5.10.2 Design View (Activity Diagram Placeholder)



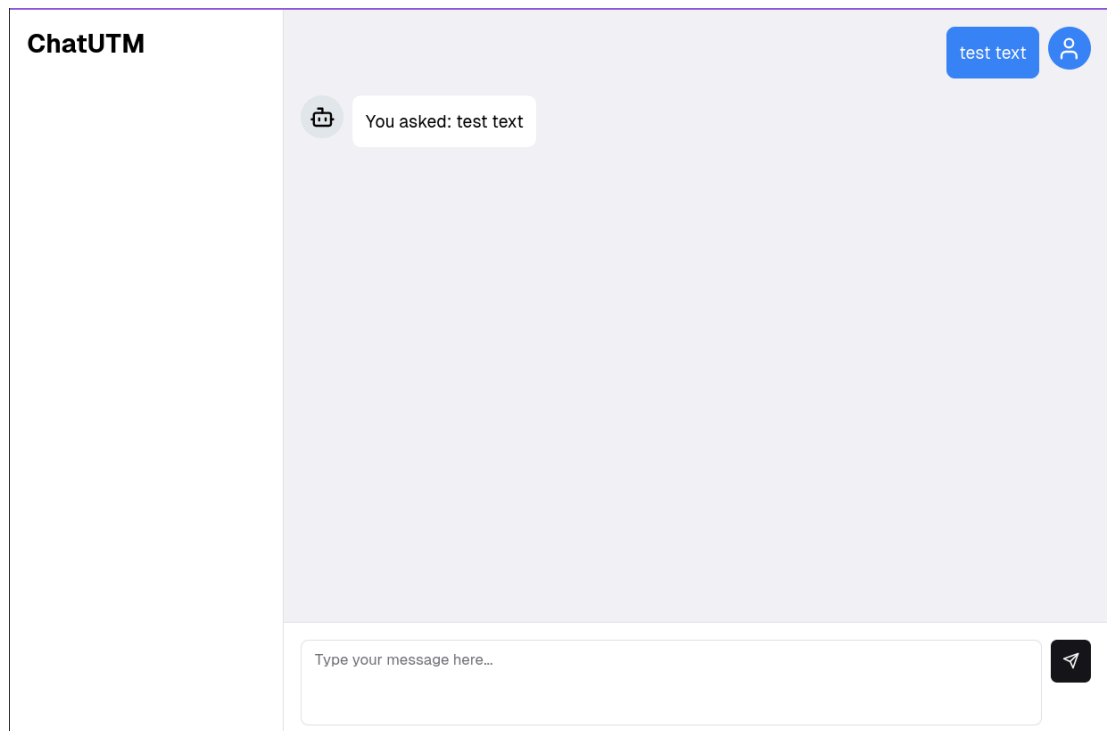
6. User Interface Design

6.1 Overview of User Interface

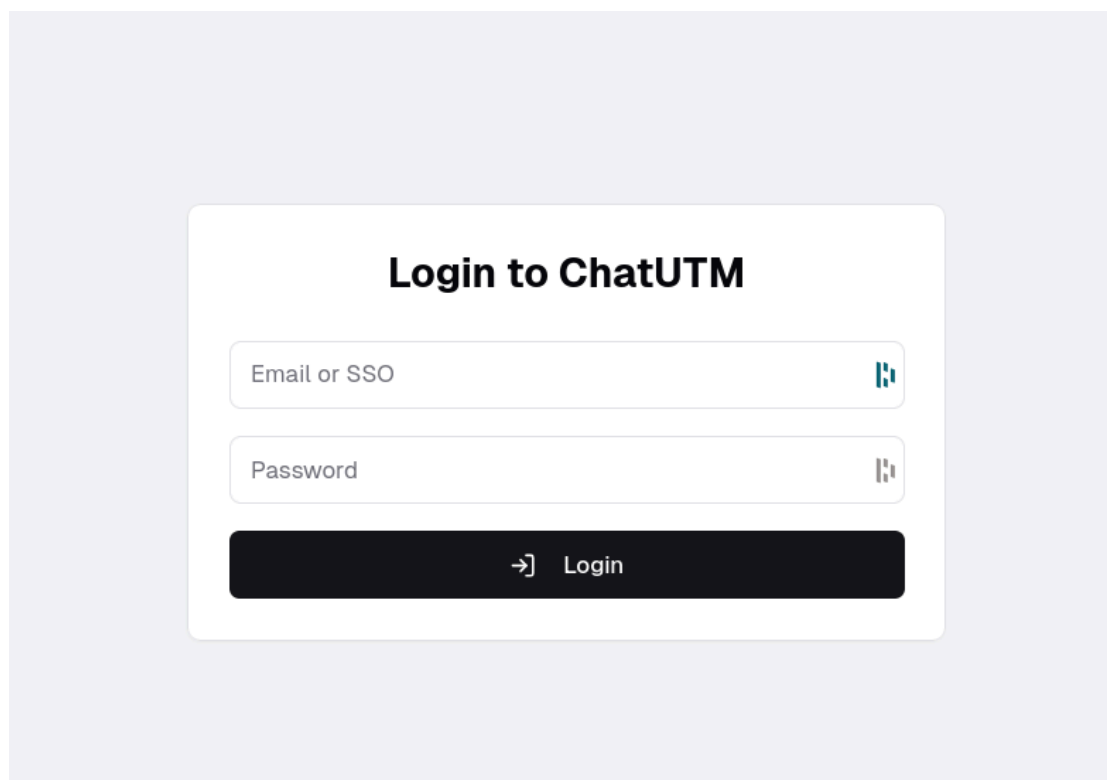
- **Web-based chat interface for real-time Q&A.**
- **Suggested questions for faster navigation.**
- **Feedback and reporting options for quality control.**

6.2 Screen Images (Placeholder for UI Screenshots)

Chat Page



Login Page



7. Appendices

- Additional diagrams, explanations, or supporting data.

Software Test Documentation (STD) for ChatUTM

Version 1.0

Prepared by: Ahmed Ghaleb

Faculty of Computing, Universiti Teknologi Malaysia

1. Revision Page

1.1 Overview

This **Software Test Documentation (STD)** describes the **testing strategy, test cases, and validation methods** to ensure that **ChatUTM** meets its **functional and non-functional requirements**. The **primary focus** is on:

- **Core functionality** (authentication, question answering, navigation).
- **Performance** (handling **1000 concurrent users**).
- **Scalability** (ensuring **system stability** under load).
- **Security** (access control via **UTM SSO authentication**).

1.2 Target Audience

This document is intended for:

- ✓ **Developers** – To execute test cases and resolve issues.
- ✓ **System Testers** – To ensure system reliability and correctness.
- ✓ **Project Stakeholders** – To review the system's test coverage and performance.

1.3 Project Team Members

Name		Role	Assigned Module
Ahmed Ghaleb		Developer	Full System Testing

1.4 Version Control History

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Ahmed Ghaleb	Initial Submission	02/01/2025
1.0	Ahmed Ghaleb	More use cases added	08/02/2025

2. Introduction

2.1 Purpose

This **Software Test Documentation (STD)** provides detailed information about:

- The **testing activities** related to **ChatUTM**.
- The **test description, expected outcomes, and validation methods**.
- Ensuring that **ChatUTM** meets the **required reliability and usability standards**.

2.2 Scope

The STD covers **functional and non-functional testing** for the following **ChatUTM features**:

- ✓ **User Authentication** (via UTM SSO).
- ✓ **AI Query Processing & Answer Retrieval**.

- ✓ **Suggested Questions & Navigation.**
- ✓ **User Feedback and Reporting Mechanisms.**
- ✓ **Multilingual Support & Language Switching.**
- ✓ **Chat History Management.**
- ✓ **Answer Exporting (Copy, PDF, Link Sharing).**

2.3 Definitions, Acronyms, and Abbreviations

Term	Definition
FAISS	Facebook AI Similarity Search – Vector database for semantic search.
NLP	Natural Language Processing – AI technique for interpreting user queries.
DeepSeek Embeddings	AI-generated embeddings for knowledge retrieval.
Docker	A containerization platform for scalable system deployment.

2.4 References

- **IEEE Std. 829-2008** – IEEE Standard for Software Test Documentation.
- **FAISS Documentation** – <https://faiss.ai>
- **DeepSeek Embeddings** – <https://deepseek.ai>

2.5 System Overview

This STD provides a **structured testing plan** for **ChatUTM**, covering **functional and performance aspects**. The document is **organized as follows**:

- **Section 3** – Test cases and execution details.
- **Section 4** – Test execution instructions.
- **Section 5** – Test results and reporting structure.

3. Test Cases, Data, and Expected Results

3.1 Functional Test Cases

Test TC001 for Module: User Authentication (UC001)

Test Case ID	Test Case Description	Created by	Version
TC001_01	Login with valid credentials	Ahmed Ghaleb	1.0
TC001_02	Login with invalid credentials	Ahmed Ghaleb	1.0

Test Case: TC001_01 – Login with Valid Credentials

No.	Prerequisites
1	User must have valid UTM SSO credentials.

No.	Test Data
-----	-----------

1	Username: student@utm.my , Password: correct-password
---	--

Test Conditions	
Step #	Step Details
1	Enter valid UTM SSO credentials
2	Click Login
3	Observe system response

Test Case: TC001_02 – Login with Invalid Credentials

No.	Prerequisites
1	User enters incorrect credentials.

No.	Test Data
1	Username: student@utm.my , Password: wrong-password

Test Conditions	
Step #	Step Details
1	Enter invalid UTM credentials
2	Click Login
3	Observe system response

3.2 Functional Test Cases for Query Processing (UC002)

Test Case ID	Feature	Test Steps	Expected Result
TC002_01	Ask and Receive Answers	Enter a question in chat	AI-generated response with references
TC002_02	Alternative Answer Retrieval	Enter a question with no exact match	System provides the closest relevant answer

3.3 Non-Functional Test Cases

Test Case ID	Feature	Test Steps	Expected Result
TC-NF01	Performance	Simulate query requests	Response within 2 sec
TC-NF02	Scalability	Simulate 1000 concurrent users	System handles load
TC-NF03	Security	Attempt login with invalid credentials	Unauthorized access is rejected

TC-NF0 4	Data Integrity	Enter multiple identical queries	System provides consistent responses
---------------------	-------------------	-------------------------------------	---

4. Test Execution Instructions

4.1 Functional Testing

- Manually execute each test case or **use automated testing scripts**.
- Record **pass/fail status** for each test.
- Document any **unexpected behavior or errors**.

4.2 Performance Testing

- Use **load-testing tools** to simulate **1000 concurrent users**.
- Measure **response time** and **system stability**.

4.3 Usability Testing

- Conduct tests with **UTM students and staff**.
 - Gather feedback on **ease of use and navigation**.
-

5. Test Results and Reporting

5.1 Test Log

- **Record all test executions**, including **pass/fail status**.

5.2 Issue Tracking

- **Log bugs or system issues** discovered during testing.

5.3 Summary Report

- Includes **performance benchmarks**, **security test results**, and **resolved issues**.
-

6. Appendices

Appendix A: Traceability Matrix

Test Case ID	Use Case ID	Sequence Diagram ID	Package ID
TC001 (Login)	UC001	SD001	P001
TC002 (Queries)	UC002	SD002	P001
TC003 (Feedback)	UC003	SD003	P002
TC004 (Navigation)	UC004	SD004	P002